

with long-term catheters have occult pyelonephritis. A randomized trial in persons with spinal cord injury who were undergoing intermittent catheterization found that relapse was more common after 3 days of therapy than after 14 days. In general, a 7- to 14-day course of antibiotics is recommended, but further studies on the optimal duration of therapy are needed.

The best strategy for prevention of CAUTI is to avoid insertion of unnecessary catheters and to remove catheters once they are no longer necessary. Quality-improvement collaboratives that have addressed technical aspects of CAUTI prevention (such as avoidance of inappropriate catheterization) as well as team communication strategies have shown the benefit of this approach in decreasing CAUTI in both acute- and long-term-care settings. Antimicrobial catheters impregnated with silver or nitrofurazone have not been shown to provide significant clinical benefit in terms of reducing rates of symptomatic UTI. Evidence is insufficient to recommend suprapubic catheters and condom catheters as alternatives to indwelling urinary catheters as a means to prevent bacteriuria. However, intermittent catheterization may be preferable to long-term indwelling urethral catheterization in certain populations (e.g., spinal cord-injured persons) to prevent both infectious and anatomic complications.

CANDIDURIA

The appearance of *Candida* in the urine is an increasingly common complication of indwelling catheterization, particularly for patients in the intensive care unit, those taking broad-spectrum antimicrobial drugs, and those with underlying diabetes mellitus. In many studies, >50% of urinary *Candida* isolates have been found to be non-*albicans* species. The clinical presentation varies from a laboratory finding without symptoms to pyelonephritis and even sepsis. Removal of the urethral catheter results in resolution of candiduria in more than one-third of asymptomatic cases. Treatment of asymptomatic patients does not appear to decrease the frequency of recurrence of candiduria. Therapy is recommended for patients who have symptomatic cystitis or pyelonephritis and for those who are at high risk for disseminated disease. High-risk patients include those with neutropenia, those who are undergoing urologic manipulation, those who are clinically unstable, and low-birth-weight infants. Fluconazole (200–400 mg/d for 7–14 days) reaches high levels in urine and is the first-line regimen for *Candida* infections of the urinary tract. Although instances of successful eradication of candiduria by some of the newer azoles and echinocandins have been reported, these agents are characterized by only low-level urinary excretion and thus are not recommended. For *Candida* isolates with high levels of resistance to fluconazole, oral flucytosine and/or parenteral amphotericin B are options. Bladder irrigation with amphotericin B generally is not recommended.

PREVENTION OF RECURRENT UTI IN WOMEN

Recurrence of uncomplicated cystitis in reproductive-age women is common, and a preventive strategy is indicated if recurrent UTIs are interfering with a patient's lifestyle. The threshold of two or more symptomatic episodes per year is not absolute; decisions about interventions should take the patient's preferences into account.

Three prophylactic strategies are available: continuous, postcoital, and patient-initiated therapy. Continuous prophylaxis and postcoital prophylaxis usually entail low doses of TMP-SMX or nitrofurantoin. These regimens are all highly effective during the period of active antibiotic intake. Typically, a prophylactic regimen is prescribed for 6 months and then discontinued, at which point the rate of recurrent UTI often returns to baseline. If bothersome infections recur, the prophylactic program can be reinstituted for a longer period. Selection of resistant strains in the fecal flora has been documented in studies of women taking prophylactic antibiotics for 12 months.

Patient-initiated therapy involves supplying the patient with materials for urine culture and with a course of antibiotics for self-medication at the first symptoms of infection. The urine culture is refrigerated and delivered to the physician's office for confirmation of the diagnosis.

When an established and reliable patient-provider relationship exists, the urine culture can be omitted as long as the symptomatic episodes respond completely to short-course therapy and are not followed by relapse.

Non-antimicrobial prevention is increasingly being studied. Lactobacillus probiotics are one appealing approach to UTI prevention, but there is a paucity of data to support this strategy. Similarly, studies of cranberry products for UTI prevention have produced mixed results. Varied dosing and product composition between studies remain an issue for providing clinical guidance.

PROGNOSIS

Cystitis is a risk factor for recurrent cystitis and pyelonephritis. ASB is common among elderly and catheterized patients but does not in itself increase the risk of death. The relationships among recurrent UTI, chronic pyelonephritis, and renal insufficiency have been widely studied. In the absence of anatomic abnormalities such as reflux, recurrent infection in children and adults does not lead to chronic pyelonephritis or to renal failure. Moreover, infection does not play a primary role in chronic interstitial nephritis; the primary etiologic factors in this condition are analgesic abuse, obstruction, reflux, and toxin exposure. In the presence of underlying renal abnormalities (particularly obstructing stones), infection as a secondary factor can accelerate renal parenchymal damage.

FURTHER READING

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Sexually Transmitted Infections: Overview and Clinical Approach

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CLASSIFICATION AND EPIDEMIOLOGY

Worldwide, most adults acquire at least one sexually transmitted infection (STI), and many remain at risk for complications. Each day, for example, more than 1 million STIs are acquired worldwide, placing many affected persons at risk for adverse reproductive health outcomes and neoplasia. Certain STIs, such as syphilis, gonorrhea, HIV infection, hepatitis B, and chancroid, often occur in highly interconnected sexual networks characterized by high rates of partner change or multiple concurrent partners. Such networks, for example, often include

persons who engage in transactional sex, men who have sex with men (MSM), and persons involved in the use of illicit drugs, particularly methamphetamine. Other STIs are distributed more evenly throughout populations. For example, chlamydial infections, genital human papillomavirus (HPV) infections, and genital herpes can spread efficiently even in relatively low-risk populations. Finally, modern technologies based on detection of nucleic acid have accelerated elucidation of the role of sexual transmission in the spread of some viruses, including Ebola virus and Zika virus, and have provided new evidence of apparent sexual transmission of several bacteria, including group C *Neisseria meningitidis* and anaerobes associated with bacterial vaginosis (BV).

In general, the product of three factors determines the initial rate of spread of any STI within a population: rate of sexual exposure of susceptible to infectious people, efficiency of transmission per exposure, and duration of infectivity of those infected. Accordingly, efforts to prevent and control STIs aim to decrease the rate of sexual exposure of susceptible to infected persons (e.g., through education and efforts to change sexual behavior norms and through control efforts aimed at reducing the proportion of the population infected); to decrease the duration of infectivity (through early diagnosis and curative or suppressive treatment); and to decrease the efficiency of transmission (through promotion of condom use and safer sexual practices, use of effective vaccines, and male medical circumcision).

In all societies, STIs rank among the most common of all infectious diseases, with at least 40 microorganisms now classified as predominantly sexually transmitted or as frequently sexually transmissible (Table 136-1). In developing countries, with three-quarters of the world's population and 90% of the world's STIs, factors such as population growth (especially in adolescent and young-adult age groups), rural-to-urban migration, wars, limited or no provision of reproductive

health services for women, and poverty create exceptional vulnerability to disease resulting from unprotected sex. During the 1990s in China, Russia, the other states of the former Soviet Union, and South Africa, internal social structures changed rapidly as borders opened to the West, unleashing enormous new epidemics of HIV infection and other STIs. Despite advances in the provision of highly effective antiretroviral therapy worldwide, HIV remains the leading cause of death in some developing countries, and HPV and hepatitis B virus (HBV) remain important causes of cervical and hepatocellular carcinoma, respectively—two of the most common (and preventable) malignancies in the developing world. Sexually transmitted herpes simplex virus (HSV) infection causes most genital ulcer disease throughout the world, and an increasing proportion of cases of genital herpes occur in developing countries with generalized HIV epidemics, where the positive-feedback loop between HSV and HIV transmission remains intractable. Despite this consistent link, randomized trials evaluating the efficacy of antiviral therapy in suppressing HSV in both HIV-uninfected and HIV-infected persons have demonstrated no protective effect against acquisition or transmission of HIV. The World Health Organization estimated that 357 million new cases of four curable STIs—gonorrhea, chlamydial infection, syphilis, and trichomoniasis—occurred annually in recent years. Up to 50% of women of reproductive age in developing countries have BV (arguably acquired sexually). All of these curable STIs have been associated with increased risk of HIV transmission or acquisition.

In the United States, the prevalence of antibody to HSV-2 began to fall in the late 1990s, especially among adolescents and young adults; the decline was presumably due to delayed sexual debut, increased condom use, and lower rates of multiple (four or more) sex partners—all well documented by the U.S. Youth Risk Behavior Surveillance System. The estimated annual incidence of HBV infection has also declined dramatically since the mid-1980s; this decrease is probably attributable to now-widespread administration of hepatitis B vaccine in infancy. Genital HPV remains the most common sexually transmitted pathogen in the United States, infecting 60% of a cohort of initially HPV-negative, sexually active Washington state college women within 5 years in a study conducted from 1990 to 2000—i.e., during the pre-HPV immunization era. Global expansion of HPV vaccine coverage among young women has already shown promise in reducing the incidence of infection with the HPV types included in the vaccines and of conditions associated with these viruses, including invasive cervical cancer.

In industrialized countries, fear of HIV infection in the mid-1980s and through the mid-2000s, coupled with widespread behavioral interventions and better-organized systems of care for the curable STIs, initially helped curb the transmission of several STDs. However, with current antiretroviral therapy, HIV has become for many a chronic disease associated with a normal life span and high quality of life. Rates of gonorrhea and syphilis remain higher in the United States than in any other Western industrialized country.

In the United States, the Centers for Disease Control and Prevention (CDC) has compiled reported rates of STIs since 1941. The incidence of reported gonorrhea peaked at 468 cases per 100,000 population in the mid-1970s and fell to a low of 98 cases per 100,000 in 2012; in 2018, the case rate was 179.1 per 100,000 persons, which is more than an 80% increase since 2009 when the number of new cases reached an all-time low. With increased testing and more sensitive tests, the incidence of reported *Chlamydia trachomatis* infection has been increasing steadily since reporting began in 1984, reaching an all-time peak of 457.6 cases per 100,000 in 2011. The incidence of primary and secondary syphilis per 100,000 peaked at 71 cases in 1946, fell rapidly to 3.9 cases in 1956, ranged from ~10 to 15 cases through 1987 (with markedly increased rates among MSM and African Americans), and then fell to a nadir of 2.1 cases in 2000–2001 (with rates falling most rapidly among heterosexual African Americans). However, since 1996, with the introduction of highly active antiretroviral therapy, gonorrhea, syphilis, and chlamydial infection have had a remarkable resurgence among MSM in North America and Europe, where outbreaks of a rare type of chlamydial infection (lymphogranuloma venereum [LGV])

TABLE 136-1 Sexually Transmitted and Sexually Transmissible Microorganisms		
BACTERIA	VIRUSES	OTHER ^a
Transmitted in Adults Predominantly by Sexual Intercourse		
<i>Neisseria gonorrhoeae</i>	HIV (types 1 and 2)	<i>Trichomonas vaginalis</i>
<i>Chlamydia trachomatis</i>	Human T-cell lymphotropic virus type 1	<i>Phthirus pubis</i>
<i>Treponema pallidum</i>	Herpes simplex virus type 2	
<i>Haemophilus ducreyi</i>	Human papillomavirus (multiple genital genotypes)	
<i>Klebsiella (Calymmatobacterium) granulomatis</i>	Hepatitis B virus ^b	
<i>Ureaplasma urealyticum</i>	Molluscum contagiosum virus	
<i>Mycoplasma genitalium</i>		
Sexual Transmission Repeatedly Described but Not Well Defined or Not the Predominant Mode		
<i>Mycoplasma hominis</i>	Cytomegalovirus	<i>Candida albicans</i>
<i>Gardnerella vaginalis</i> and other vaginal bacteria	Human T-cell lymphotropic virus type 2	<i>Sarcoptes scabiei</i>
Group B <i>Streptococcus</i>	Hepatitis C virus	
<i>Mobiluncus</i> spp.	(?) Hepatitis D virus	
<i>Helicobacter cinaedi</i>	Herpes simplex virus type 1	
<i>Helicobacter fennelliae</i>	Zika virus	
Anaerobes associated with bacterial vaginosis	Ebola virus	
<i>Leptotrichia/Sneathia</i>	(?) Epstein-Barr virus	
Group C <i>Neisseria meningitidis</i>	Human herpesvirus type 8	
Transmitted by Sexual Contact Involving Oral–Fecal Exposure; of Declining Importance in Men Who Have Sex with Men		
<i>Shigella</i> spp.	Hepatitis A virus	<i>Giardia lamblia</i>
<i>Campylobacter</i> spp.		<i>Entamoeba histolytica</i>

^aIncludes protozoa, ectoparasites, and fungi. ^bAmong U.S. patients for whom a risk factor can be ascertained, most hepatitis B virus infections are transmitted sexually.

1080 that had virtually disappeared during the AIDS era have occurred. In 2018, ~75% of primary and secondary syphilis cases reported to the CDC were in MSM, but incidence has also increased in women with a concomitant increase in congenital syphilis. Moreover, the uptake of oral pre-exposure prophylaxis for HIV-1 acquisition has increased among MSM since its initial approval for this purpose in 2012 and has been associated with reports of reduced condom-use frequency and concomitantly increased STI acquisition. These developments have resulted in a soaring incidence of STIs, with increasing co-infection with HIV and other sexually transmitted pathogens (particularly *Treponema pallidum*, the cause of syphilis; and *Neisseria gonorrhoeae*, the cause of gonorrhea), primarily among MSM.

MANAGEMENT OF COMMON SEXUALLY TRANSMITTED DISEASE (STD) SYNDROMES

Although other chapters discuss management of specific STIs, most patients are managed (at least initially) on the basis of presenting symptoms and signs and associated risk factors, even in industrialized countries. **Table 136-2** lists some of the most common clinical STD syndromes and their microbial etiologies. Strategies for their management are outlined below. **Chapters 201 and 202 address the management of infections with human retroviruses.**

STD care and management begin with risk assessment and proceed to clinical assessment, diagnostic testing or screening, treatment, and prevention. Risk assessment guides detection and interpretation of symptoms that could denote an STD; decisions on screening or prophylactic/preventive treatment; risk reduction counseling and intervention (e.g., hepatitis B vaccination); treatment of partners of patients with known infections; and behavioral risk reduction by the patient. Consideration of routine demographic data (e.g., gender, age, area of residence) is a simple first step in this risk assessment. For example, national guidelines strongly recommend routine screening of sexually active females ≤25 years of age for *C. trachomatis* infection. **Table 136-3** provides a set of 11 STD/HIV risk-assessment questions that clinicians can pose verbally or that health care systems can adapt (with yes/no responses) into a routine self-administered questionnaire. The initial framing statement gives permission to discuss topics that may be difficult for the patient to disclose.

Risk assessment is followed by clinical assessment (elicitation of information on specific current symptoms and signs of STDs). Confirmatory diagnostic tests (for persons with symptoms or signs) or screening tests (for those without symptoms or signs) may involve microscopic examination, culture, nucleic acid amplification tests (NAATs), or serology. Initial syndrome-based treatment should cover the most likely causes. For certain syndromes, results of rapid tests can narrow the spectrum of this initial therapy (e.g., pH of vaginal fluid for women with vaginal discharge, Gram's stain of urethral discharge for men with urethral discharge, rapid plasma reagin test for genital ulcer to assess the probability of syphilis). After the institution of treatment, STD management proceeds to the "4 Cs" of prevention and control: contact tracing (see "Prevention and Control of STIs," below), ensuring compliance with therapy, and counseling on risk reduction, including condom promotion and provision as well as motivational interviewing for risk reduction.

Consistent with current guidelines, all adults should be screened for infection with HIV-1 at least once, and more frequently if they are at elevated risk for acquisition of this infection.

■ URETHRITIS IN MEN

Urethritis in men produces urethral discharge, dysuria, or both, usually without frequency of urination. Causes include *N. gonorrhoeae*, *C. trachomatis*, *Mycoplasma genitalium*, *Ureaplasma urealyticum*, *Trichomonas vaginalis*, HSV, and (rarely) adenovirus.

Until recently, *C. trachomatis* caused ~30–40% of cases of nongonococcal urethritis (NGU), particularly in heterosexual men; however, the proportion of cases due to this organism has probably declined in some populations served by effective chlamydial control programs, and older men with urethritis appear less likely to have chlamydial infection. HSV and *T. vaginalis* each cause a small proportion of NGU cases in the United States. Recently, multiple studies have consistently implicated

TABLE 136-2 Major Sexually Transmitted Disease Syndromes and Sexually Transmitted Microbial Etiologies

SYNDROME	SEXUALLY TRANSMITTED MICROBIAL ETIOLOGIES
AIDS	HIV types 1 and 2
Urethritis: males	<i>Neisseria gonorrhoeae</i> , <i>Chlamydia trachomatis</i> , <i>Mycoplasma genitalium</i> , <i>Ureaplasma urealyticum</i> (subspecies <i>urealyticum</i>), <i>Trichomonas vaginalis</i> , HSV, some anaerobic bacteria, <i>Leptotrichia/Sneathia</i>
Epididymitis	<i>C. trachomatis</i> , <i>N. gonorrhoeae</i> , and (in older men or men who have sex with men) coliform bacteria
Lower genital tract infections: females	
Cystitis/urethritis	<i>C. trachomatis</i> , <i>N. gonorrhoeae</i> , HSV
Mucopurulent cervicitis	<i>C. trachomatis</i> , <i>N. gonorrhoeae</i> , <i>M. genitalium</i>
Vulvitis	<i>Candida albicans</i> , HSV
Bartholinitis	<i>C. albicans</i> , <i>T. vaginalis</i>
Vulvovaginitis	<i>C. albicans</i> , <i>T. vaginalis</i>
BV	BV-associated bacteria (see text)
Acute pelvic inflammatory disease	<i>N. gonorrhoeae</i> , <i>C. trachomatis</i> , BV-associated bacteria, <i>M. genitalium</i> , group B streptococci
Infertility	<i>N. gonorrhoeae</i> , <i>C. trachomatis</i> , BV-associated bacteria
Ulcerative lesions of the genitalia	HSV-1, HSV-2, <i>Treponema pallidum</i> , <i>Haemophilus ducreyi</i> , <i>C. trachomatis</i> (LGV strains), <i>Klebsiella (Calymmatobacterium) granulomatis</i>
Complications of pregnancy/puerperium	Several pathogens implicated
Intestinal infections	
Proctitis	<i>C. trachomatis</i> , <i>N. gonorrhoeae</i> , HSV, <i>T. pallidum</i>
Proctocolitis or enterocolitis	<i>Campylobacter</i> spp., <i>Shigella</i> spp., <i>Entamoeba histolytica</i> , <i>Helicobacter</i> spp., other enteric pathogens
Enteritis	<i>Giardia lamblia</i>
Acute arthritis with urogenital infection or viremia	<i>N. gonorrhoeae</i> (e.g., DGI), <i>C. trachomatis</i> (e.g., reactive arthritis), HBV
Genital and anal warts	HPV (30 genital types)
Mononucleosis syndrome	CMV, HIV, EBV
Hepatitis	Hepatitis viruses, <i>T. pallidum</i> , CMV, EBV
Neoplasias	
Squamous cell dysplasias and cancers of the cervix, anus, vulva, vagina, or penis	HPV (especially types 16, 18, 31, 45)
Kaposi's sarcoma, body-cavity lymphomas	HHV-8
T-cell leukemia	HTLV-1
Hepatocellular carcinoma	HBV
Tropical spastic paraparesis	HTLV-1
Scabies	<i>Sarcoptes scabiei</i>
Pubic lice	<i>Phthirus pubis</i>

Abbreviations: BV, bacterial vaginosis; CMV, cytomegalovirus; DGI, disseminated gonococcal infection; EBV, Epstein-Barr virus; HBV, hepatitis B virus; HHV-8, human herpesvirus type 8; HPV, human papillomavirus; HSV, herpes simplex virus; HTLV, human T-cell lymphotropic virus; LGV, lymphogranuloma venereum.

M. genitalium as a probable cause of many *Chlamydia*-negative cases. Fewer studies than in the past have implicated *Ureaplasma*; the ureaplasmas have been differentiated into *U. urealyticum* and *Ureaplasma parvum*, and a few studies suggest that *U. urealyticum*—but not *U. parvum*—is associated with NGU; for this reason, neither testing nor presumptive treatment for ureaplasmas in the setting of urethritis is recommended. Coliform bacteria can cause urethritis in men who

TABLE 136-3 Eleven-Question Sexually Transmitted Disease (STD)/HIV Risk Assessment

Framing Statement

In order to provide the best care for you today and to understand your risk for certain infections, it is necessary for us to talk about your sexual behavior.

Screening Questions

- (1) Do you have any reason to think you might have a sexually transmitted infection? If so, what reason?
- (2) For all adolescents <18 years old: Have you begun having any kind of sex yet?

STD History

- (3) Have you ever had any sexually transmitted infections or any genital infections? If so, which ones?

Sexual Preference

- (4) Have you had sex with men, women, or both?

Injection Drug Use

- (5) Have you ever injected yourself ("shot up") with drugs? (If yes, have you ever shared needles or injection equipment?)
- (6) Have you ever had sex with a gay or bisexual man or with anyone who had ever injected drugs?

Characteristics of Partner(s)

- (7) Has your sex partner had any sexually transmitted infections? If so, which ones?
- (8) Has your sex partner had other sex partners during the time you've been together?

STD Symptoms Checklist

- (9) Have you recently developed any of these symptoms?

For Men

- (a) Discharge of pus (drip) from the penis
- (b) Genital sores (ulcers) or rash

For Women

- (a) Abnormal vaginal discharge (increased amount, abnormal odor, abnormal yellow color)
- (b) Genital sores (ulcers), rash, or itching

Sexual Practices, Past 2 Months (for patients answering yes to any of the above questions, to guide examination and testing)

- (10) Now I'd like to ask what parts of your body may have been sexually exposed to an STD (e.g., your penis, mouth, vagina, anus).

Query About Interest in STD Screening Tests (for patients answering no to all of the above questions)

- (11) Would you like to be tested for HIV or any other STDs today? (If yes, clinician can explore which STD and why.)

Source: Adapted from JR Curtis, KK Holmes, in KK Holmes et al (eds): *Sexually Transmitted Diseases*, 4th ed. New York, McGraw-Hill, 2008.

practice insertive anal intercourse. More recently, anaerobic bacteria that are characteristically involved in BV, especially *Leptotrichia/Sneathia* species, have occasionally been associated with urethritis in heterosexual men. Recommendations for the initial diagnosis of urethritis in men currently include specific tests only for *N. gonorrhoeae* and *C. trachomatis*; they do not yet include testing for *M. genitalium*, although a NAAT is now commercially available for the latter.

APPROACH TO THE PATIENT

Urethritis in Men

The following summarizes the approach to the male patient with suspected urethritis:

1. **Establish the presence of urethritis.** If proximal-to-distal "milk-ing" of the urethra does not express a purulent or mucopurulent discharge, even after the patient has not voided for several hours (or preferably overnight), a Gram's-stained smear of an anterior urethral specimen obtained by passage of a small urethrogenital swab 2–3 cm into the urethra usually reveals ≥ 2 neutrophils per

1000 $\times$ field when urethritis is present; in gonococcal infection, such a smear usually reveals gram-negative intracellular diplococci as well. Alternatively, the centrifuged sediment of the first 20–30 mL of voided urine—ideally collected as the first morning specimen—can be examined for inflammatory cells, either by microscopy showing ≥ 10 leukocytes per high-power field or by the leukocyte esterase test. Patients with symptoms who lack objective evidence of urethritis generally do not benefit from repeated courses of antibiotics, and other etiologies of such symptoms may be considered.

2. **Evaluate for complications or alternative diagnoses.** A brief history and examination can exclude epididymitis and systemic complications, such as disseminated gonococcal infection (DGI) and reactive arthritis. Although digital examination of the prostate gland seldom contributes to the evaluation of sexually active young men with urethritis, men with dysuria who lack evidence of urethritis as well as sexually inactive men with urethritis should undergo prostate palpation, urinalysis, and urine culture to exclude bacterial prostatitis and cystitis.
3. **Evaluate for gonococcal and chlamydial infection.** An absence of typical gram-negative diplococci on Gram's-stained smear of urethral exudate containing inflammatory cells warrants a preliminary diagnosis of NGU, as this test is 98% sensitive for the diagnosis of gonococcal urethral infection. However, most men with symptoms and/or signs of urethritis are simultaneously assessed for infection with *N. gonorrhoeae* and *C. trachomatis* by NAATs of first-catch urine. The urine specimen tested should consist of the first 10–15 mL of the stream, and if possible, patients should not have voided for the prior 2 h. Culture or NAAT for *N. gonorrhoeae* may yield positive results even when Gram's staining is negative; certain strains of *N. gonorrhoeae* can result in negative urethral Gram's stains in up to 30% of cases of urethral infection. Results of tests for gonococcal and chlamydial infection predict the patient's prognosis (with greater risk for recurrent NGU if neither chlamydiae nor gonococci are found than if either is detected) and can guide both the counseling given to the patient and the management of the patient's sexual partner(s).
4. **Treat urethritis promptly while test results are pending.**

TREATMENT

Urethritis in Men

Table 136-4 summarizes the steps in management of urethral discharge and/or dysuria in sexually active men.

In practice, if Gram's stain does not reveal gonococci, urethritis is treated with a regimen effective for NGU, such as azithromycin or doxycycline. Both are effective. Although azithromycin has been more effective than doxycycline for *M. genitalium* infection, the efficacy of azithromycin for treatment of *M. genitalium* is rapidly declining. Alternatives include moxifloxacin and pristinamycin, a streptogramin antibiotic available in some countries. If gonococci are demonstrated by Gram's stain or if no diagnostic tests are performed to exclude gonorrhea definitively, treatment should include parenteral cephalosporin therapy for gonorrhea (Chap. 156). Azithromycin is effective for treating *C. trachomatis* infection, which can cause urethral co-infection in men with gonococcal urethritis. Sexual partners with contact to the index patient in the past 60 days should also be tested for gonorrhea and chlamydial infection. Regardless of whether they are tested for these infections, however, they should receive the same regimen given to the male index case. Patients with confirmed persistence or recurrence of urethritis after treatment should be re-treated with the initial regimen if they did not comply with the original treatment or were reexposed to an untreated partner. Most persistent urethritis is due to *M. genitalium*, and prompt diagnostic testing and/or treatment for *M. genitalium* is recommended.

National and international guidelines do exist for treatment of gonococcal urethritis, typically with ceftriaxone. However,

TABLE 136-4 Management of Urethral Discharge in Men

USUAL CAUSES	USUAL INITIAL EVALUATION
<i>Chlamydia trachomatis</i>	Demonstration of urethral discharge or pyuria
<i>Neisseria gonorrhoeae</i>	Exclusion of local or systemic complications
<i>Mycoplasma genitalium</i>	Urethral Gram's stain to confirm urethritis, detect gram-negative diplococci
<i>Ureaplasma urealyticum</i>	Test for <i>N. gonorrhoeae</i> , <i>C. trachomatis</i>
<i>Trichomonas vaginalis</i>	
Herpes simplex virus	

Initial Treatment for Patient and Partners

Treat gonorrhea (unless excluded):

Ceftriaxone (500 mg IM^a)

Management of Recurrence

Confirm objective evidence of urethritis. If patient was reexposed to untreated or new partner, repeat treatment of patient and partner.

If patient was not reexposed, consider infection with *T. vaginalis*^b or antibiotic-resistant *M. genitalium*^c, and treat accordingly (metronidazole for trichomoniasis; azithromycin for *M. genitalium* followed by moxifloxacin if needed).

^aNeither oral cephalosporins nor fluoroquinolones are recommended for treatment of gonorrhea in the United States because of the emergence of increasing fluoroquinolone resistance in *N. gonorrhoeae*, especially (but not only) among men who have sex with men, and the decreasing susceptibility of a still-small proportion of gonococci to ceftriaxone (Fig. 136-1). Updates on the emergence of antimicrobial resistance in *N. gonorrhoeae* can be obtained from the Centers for Disease Control and Prevention at <http://www.cdc.gov/std>. ^bIn men, the diagnosis of *T. vaginalis* infection requires nucleic acid amplification testing of a urethral swab specimen obtained before voiding. ^c*M. genitalium* is often resistant to doxycycline and azithromycin but is usually susceptible to the fluoroquinolone moxifloxacin. Moxifloxacin can be considered for treatment of refractory nongonococcal, nonchlamydial urethritis.

consensus is still lacking on treatment of urethritis that persists after treatment and cure of gonorrhea. Ideally, the approach would involve testing for potential causes of persistent urethritis (e.g., *M. genitalium*) and antimicrobial susceptibility testing in settings and populations where antimicrobial resistance is emerging. Currently, assays are available that can detect *M. genitalium*, and some experts believe it is time to integrate such testing into STD care. If *M. genitalium* is detected, the persistent urethritis can be treated with azithromycin or moxifloxacin in light of local patterns of antimicrobial susceptibility.

In heterosexual men with a high likelihood of exposure to trichomoniasis, an intraurethral swab specimen and a first-voided urine sample should be tested for *T. vaginalis* using NAAT. Presumptive treatment with metronidazole or tinidazole (2 g by mouth in a single dose) should be given. For MSM, trichomoniasis is unlikely,

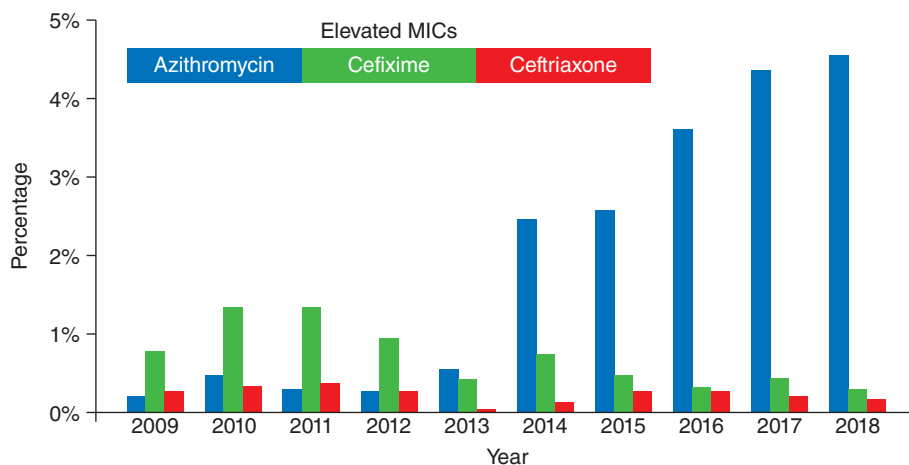
and consideration of a course of moxifloxacin is warranted. Because MSM also have the highest prevalence rates of antimicrobial-resistant *N. gonorrhoeae*, this possibility, even if apparently ruled out at the initial presentation, should be kept in mind.

EPIDIDYMITIS

Acute epididymitis, almost always unilateral, produces pain, swelling, and tenderness of the epididymis, with or without symptoms or signs of urethritis. This condition must be differentiated from testicular torsion, tumor, and trauma. Torsion, a surgical emergency, usually occurs in the second or third decade of life and produces a sudden onset of pain, elevation of the testicle within the scrotal sac, rotation of the epididymis from a posterior to an anterior position, and absence of blood flow on Doppler ultrasound. Persistence of symptoms after a course of therapy for epididymitis suggests the possibility of testicular tumor or of a chronic granulomatous disease, such as tuberculosis. In sexually active men under age 35, acute epididymitis is caused most frequently by *C. trachomatis* and less commonly by *N. gonorrhoeae* and is usually associated with overt or subclinical urethritis. Acute epididymitis occurring in older men or following urinary tract instrumentation is usually caused by urinary pathogens. These older men usually have no urethritis but do have bacteriuria. Similarly, epididymitis in MSM who have practiced insertive rectal intercourse is often caused by Enterobacteriaceae.

TREATMENT**Epididymitis**

Ceftriaxone (500 mg as a single dose IM) followed by doxycycline (100 mg by mouth twice daily for 10 days) constitutes effective treatment for epididymitis caused by *N. gonorrhoeae* or *C. trachomatis*. Neither oral cephalosporins nor fluoroquinolones are recommended for treatment of gonorrhea in the United States because of resistance in *N. gonorrhoeae*, especially (but not only) among MSM (Fig. 136-1). Given rapidly escalating rates of resistance in *N. gonorrhoeae* to azithromycin, this antibiotic is no longer recommended as co-therapy with a parenteral ceftriaxone for gonorrhea. When infection with Enterobacteriaceae is suspected, oral levofloxacin (500 mg once daily for 10 days) or ofloxacin (300 mg twice daily for 10 days) is effective for syndrome-based initial treatment of epididymitis; however, because this regimen is not effective against gonococcal or chlamydial infection, it should be combined with effective therapy for possible gonococcal or chlamydial infection of the epididymis unless bacteriuria with Enterobacteriaceae is confirmed.



NOTE: Elevated MIC = Azithromycin: ≥ 2.0 $\mu\text{g/mL}$; Cefixime: ≥ 0.25 $\mu\text{g/mL}$; Ceftriaxone: ≥ 0.125 $\mu\text{g/mL}$.

FIGURE 136-1 Percentage of *Neisseria gonorrhoeae* isolates with elevated minimal inhibitory concentrations (MICs) to azithromycin, ceftriaxone, and cefixime, United States, 2009–2018. (From the Centers for Disease Control and Prevention: Gonococcal Isolate Surveillance Project [GISP], 2018.)

■ URETHRITIS AND THE URETHRAL SYNDROME IN WOMEN

C. trachomatis, *N. gonorrhoeae*, and occasionally HSV cause symptomatic urethritis—known as the *urethral syndrome* in women—that is characterized by “internal” dysuria (usually without urinary urgency or frequency), pyuria, and an absence of *Escherichia coli* and other uropathogens at counts of $\geq 10^2$ /mL in urine. In contrast, the dysuria associated with vulvar herpes or vulvovaginal candidiasis (and perhaps with trichomoniasis) is often described as “external,” being caused by painful contact of urine with the inflamed or ulcerated labia or introitus. Acute onset, association with urinary urgency or frequency, hematuria, or suprapubic bladder tenderness suggests bacterial cystitis. Among women with symptoms of acute bacterial cystitis, costovertebral pain and tenderness or fever suggest acute pyelonephritis. **The management of bacterial urinary tract infection (UTI) is discussed in Chap. 135.**

Signs of vulvovaginitis, coupled with symptoms of external dysuria, suggest vulvar infection (e.g., with HSV or *Candida albicans*). Among dysuric women without signs of vulvovaginitis, bacterial UTI must be differentiated from the urethral syndrome by assessment of risk, evaluation of the pattern of symptoms and signs, and specific microbiologic testing. An STI etiology of the urethral syndrome is suggested by young age, more than one current sexual partner, a new partner within the past month, a partner with urethritis, or coexisting mucopurulent cervicitis (see below). The finding of a single urinary pathogen, such as *E. coli* or *Staphylococcus saprophyticus*, at a concentration of $\geq 10^2$ /mL in a properly collected specimen of midstream urine from a dysuric woman with pyuria indicates probable bacterial UTI, whereas pyuria with $<10^2$ conventional uropathogens per milliliter of urine (“sterile” pyuria) suggests acute urethral syndrome due to *C. trachomatis* or *N. gonorrhoeae*. Gonorrhea and chlamydial infection should be sought by specific tests (e.g., NAATs of vaginal secretions collected with a swab). Among dysuric women with sterile pyuria caused by infection with *N. gonorrhoeae* or *C. trachomatis*, appropriate treatment alleviates dysuria. The role of *M. genitalium* in the urethral syndrome in women remains undefined.

■ VULVOVAGINAL INFECTIONS

Abnormal Vaginal Discharge If directly questioned about vaginal discharge during routine health checkups, many women acknowledge having nonspecific symptoms of vaginal discharge that do not correlate with objective signs of inflammation or with actual infection. However, unsolicited reporting of abnormal vaginal discharge often denotes BV or trichomoniasis. Specifically, an abnormally increased amount or an abnormal odor of the discharge is associated with one or both of these conditions. Cervical infection with *N. gonorrhoeae* or *C. trachomatis* does not often cause an increased amount or abnormal odor of discharge; however, when these pathogens cause cervicitis, they—like *T. vaginalis*—often result in an increased number of neutrophils in vaginal fluid, which thus takes on a yellow color. Vulvar conditions such as genital herpes or vulvovaginal candidiasis can cause vulvar pruritus, burning, irritation, or lesions as well as external dysuria (as urine passes over the inflamed vulva or areas of epithelial disruption) or vulvar dyspareunia.

Certain vulvovaginal infections may have serious sequelae. Trichomoniasis, BV, and vulvovaginal candidiasis have all been associated with increased risk of acquisition of HIV infection; BV promotes HIV transmission from HIV-infected women to their male sex partners. Vaginal trichomoniasis and BV early in pregnancy independently predict premature onset of labor. BV can also lead to anaerobic bacterial infection of the endometrium and salpinges. Vaginitis may be an early and prominent feature of toxic shock syndrome, and recurrent or chronic vulvovaginal candidiasis develops with increased frequency among women who have systemic illnesses, such as diabetes mellitus or HIV-related immunosuppression (although only a very small proportion of women with recurrent vulvovaginal candidiasis in industrialized countries actually have a serious predisposing illness).

Thus, vulvovaginal symptoms or signs warrant careful evaluation, including speculum and pelvic examination, diagnostic testing, and appropriate therapy specific for the infection identified. Unfortunately, clinicians do not always perform the tests required to establish the

cause of such symptoms. Further, self-diagnosis of a specific type of infection—including vulvovaginal candidiasis—is often incorrect. The diagnosis and treatment of the three most common types of vaginal infection are summarized in **Table 136-5**.

Inspection of the vulva and perineum may reveal tender genital ulcerations or fissures (typically due to HSV infection or vulvovaginal candidiasis) or discharge visible at the introitus before insertion of a speculum (suggestive of BV or trichomoniasis). Speculum examination permits the clinician to discern whether the discharge appears abnormal and whether it emanates from the cervical os (mucoid and, if abnormal, yellow) or from the vagina (not mucoid, since the vaginal epithelium does not produce mucus). Symptoms or signs of abnormal vaginal discharge should prompt testing of vaginal fluid for pH, for a fishy odor when mixed with 10% KOH, and for certain microscopic features when mixed with saline (motile trichomonads and/or “clue cells”) and with 10% KOH (pseudohyphae or hyphae indicative of vulvovaginal candidiasis). Additional objective laboratory tests, described below, are useful for establishing the cause of abnormal vaginal discharge. Gram’s staining of vaginal fluid can be used to characterize the vaginal bacteria using the Nugent score but is used primarily for research purposes and requires familiarity with the morphotypes and scale involved. Of note, NAATs that characterize relative concentrations of BV-associated bacteria and certain *Lactobacillus* species are now available and offer comparable performance to clinical diagnostic criteria.

TREATMENT

Vaginal Discharge

Patterns of treatment for abnormal vaginal discharge vary widely. In developing countries, where clinics or pharmacies often dispense treatment based on symptoms alone without examination or testing, oral treatment with metronidazole—particularly with a 7-day regimen—provides reasonable coverage against both trichomoniasis and BV, the usual causes of symptoms of vaginal discharge. Metronidazole treatment of sex partners prevents reinfection of women with *T. vaginalis*, although it does not help prevent the recurrence of BV. Guidelines for syndromic management promulgated by the World Health Organization suggest consideration of treatment for cervical infection and for trichomoniasis, BV, and vulvovaginal candidiasis in women with symptoms of abnormal vaginal discharge. However, it is important to note that the majority of chlamydial and gonococcal cervical infections produce no symptoms.

In industrialized countries, clinicians treating symptoms and signs of abnormal vaginal discharge should, at a minimum, differentiate between BV and trichomoniasis because optimal management of patients and partners differs for these two conditions.

Vaginal Trichomoniasis (See also Chap. 229) Symptomatic trichomoniasis characteristically produces a profuse, yellow, purulent, homogeneous vaginal discharge and vulvar irritation, sometimes with visible inflammation of the vaginal and vulvar epithelium and petechial lesions on the cervix (the so-called strawberry cervix, best visualized by colposcopy). The pH of vaginal fluid—normally <4.7 —usually rises to ≥ 5 . Microscopic examination of vaginal discharge mixed with saline reveals motile trichomonads in most culture-positive cases. However, saline microscopy probably detects only one-half of all cases, and, especially in the absence of symptoms or signs, culture or NAAT is usually required for detection of the organism. NAAT for *T. vaginalis* is more sensitive than culture. Treatment of asymptomatic as well as symptomatic cases reduces rates of transmission and prevents later development of symptoms.

TREATMENT

Vaginal Trichomoniasis

Only nitroimidazoles (e.g., metronidazole and tinidazole) consistently cure trichomoniasis. A single 2-g oral dose of metronidazole has been the standard treatment for decades, but it is less effective

TABLE 136-5 Diagnostic Features and Management of Vaginal Infection

FEATURE	NORMAL VAGINAL EXAMINATION	VULVOVAGINAL CANDIDIASIS	TRICHOMONAL VAGINITIS	BACTERIAL VAGINOSIS (BV)
Etiology	Uninfected; lactobacilli predominant	<i>Candida albicans</i>	<i>Trichomonas vaginalis</i>	Associated with <i>Gardnerella vaginalis</i> , various anaerobic bacteria, and mycoplasmas
Typical symptoms	None	Vulvar itching and/or irritation	Profuse discharge; vulvar itching	Malodorous, slightly increased discharge
Discharge				
Amount	Variable; usually scant	Scant	Often profuse	Moderate
Color ^a	Clear or translucent	White	White or yellow	White or gray
Consistency	Nonhomogeneous, flocculent	Clumped; adherent plaques	Homogeneous	Homogeneous, low viscosity; uniformly coats vaginal walls
Inflammation of vulvar or vaginal epithelium	None	Erythema of vaginal epithelium, introitus; vulvar dermatitis, fissures common	Erythema of vaginal and vulvar epithelium; colpitis macularis	None
pH of vaginal fluid ^b	Usually ≤ 4.5	Usually ≤ 4.5	Usually ≥ 5	Usually > 4.5
Amine ("fishy") odor with 10% KOH	None	None	May be present	Present
Microscopy ^c	Normal epithelial cells; lactobacilli predominant	Leukocytes, epithelial cells; mycelia or pseudomycelia in up to 80% of <i>C. albicans</i> culture-positive persons with typical symptoms	Leukocytes; motile trichomonads seen in 80–90% of symptomatic patients, less often in the absence of symptoms	Clue cells; few leukocytes; no lactobacilli or only a few outnumbered by profuse mixed microbiota, nearly always including <i>G. vaginalis</i> plus anaerobic species on Gram's stain (Nugent's score ≥ 7)
Other laboratory findings		Isolation of <i>Candida</i> spp.	Isolation of <i>T. vaginalis</i> or positive NAAT ^d	Diagnosis of BV by NAAT ^d
Usual treatment	None	Azole cream, tablet, or suppository—e.g., miconazole (100-mg vaginal suppository) or clotrimazole (100-mg vaginal tablet) once daily for 7 days Fluconazole, 150 mg orally (single dose)	Metronidazole or tinidazole, 2 g orally (single dose) Metronidazole, 500 mg PO bid for 7 days	Metronidazole, 500 mg PO bid for 7 days Metronidazole gel, 0.75%, one applicator (5 g) intravaginally once daily for 5 days Clindamycin, 2% cream, one full applicator vaginally each night for 7 days
Usual management of sexual partner	None	None; topical treatment if candidal dermatitis of penis is detected	Examination for sexually transmitted infection; treatment with metronidazole, 2 g PO (single dose)	None

^aColor of discharge is best determined by examination against the white background of a swab. ^bA pH determination is not useful if blood is present or if the test is performed on endocervical secretions. ^cTo detect fungal elements, vaginal fluid is digested with 10% KOH prior to microscopic examination; to examine for other features, fluid is mixed (1:1) with physiologic saline. Gram's stain is also excellent for detecting yeasts (less predictive of vulvovaginitis) and pseudomycelia or mycelia (strongly predictive of vulvovaginitis) and for distinguishing normal flora from the mixed flora seen in bacterial vaginosis, but it is less sensitive than the saline preparation for detection of *T. vaginalis*. ^dNAAT, nucleic acid amplification test (where available). NAAT for diagnosis of BV typically tests for combinations of BV-associated bacteria and absence of *Lactobacillus* species.

than a weeklong course; the latter is preferred. Tinidazole has a longer half-life than metronidazole, causes fewer gastrointestinal symptoms, and may be useful in treating trichomoniasis that fails to respond to metronidazole. Treatment of sexual partners, facilitated by dispensing metronidazole to the female patient to give to her partner(s), significantly reduces both the risk of reinfection and the reservoir of infection; treating partners is the standard of care. Intravaginal treatment with 0.75% metronidazole gel is not reliable for vaginal trichomoniasis. Thus, systemic use of metronidazole is still recommended throughout pregnancy for treatment of trichomoniasis. In a large randomized trial, metronidazole treatment of trichomoniasis during pregnancy was associated with an increased frequency of perinatal morbidity. However, most studies, including randomized controlled trials, have shown no adverse effects of metronidazole use during pregnancy on preterm birth or birth defects.

Bacterial Vaginosis BV is a syndrome characterized by symptoms of vaginal malodor and increased white-gray discharge, which appears homogeneous, is low in viscosity, and uniformly covers the vaginal mucosa. BV has been associated with an increased risk of acquiring several other genital infections, including those caused by HIV, *C. trachomatis*, and *N. gonorrhoeae*. Other possible risk factors include recent unprotected vaginal intercourse, having a female sex partner, and vaginal douching. Although bacteria associated with BV have

been detected under the foreskin of uncircumcised men and have been associated with urethritis, metronidazole treatment of male partners has not reduced the rate of recurrence of BV among affected women.

Among women with BV, culture of vaginal fluid has shown markedly increased prevalences and concentrations of *Gardnerella vaginalis*, *Mycoplasma hominis*, and several anaerobic bacteria (e.g., *Mobiluncus*, *Prevotella* [formerly *Bacteroides*], and some *Peptostreptococcus* species) as well as an absence of hydrogen peroxide-producing *Lactobacillus* species that constitute most of the normal vaginal microbiota and help protect against cervical and vaginal infections. Broad-range polymerase chain reaction (PCR) amplification of 16S rDNA in vaginal fluid, with subsequent identification of specific bacterial species by various methods, has documented even greater bacterial diversity, including several unique species not previously identified in culture (Fig. 136-2) and *Atopobium vaginae*, an organism that is strongly associated with BV and is resistant to metronidazole. Other genera newly implicated in BV include *Megasphaera*, *Leptotrichia*, *Eggerthella*, and *Dialister*.

BV is conventionally diagnosed clinically with the Amsel criteria, which include any three of the following four clinical abnormalities: (1) objective signs of increased white homogeneous vaginal discharge; (2) a vaginal discharge pH of > 4.5 ; (3) liberation of a distinct fishy odor (attributable to volatile amines such as trimethylamine) immediately after vaginal secretions are mixed with a 10% solution of KOH; and (4) microscopic demonstration of "clue cells" (vaginal epithelial cells

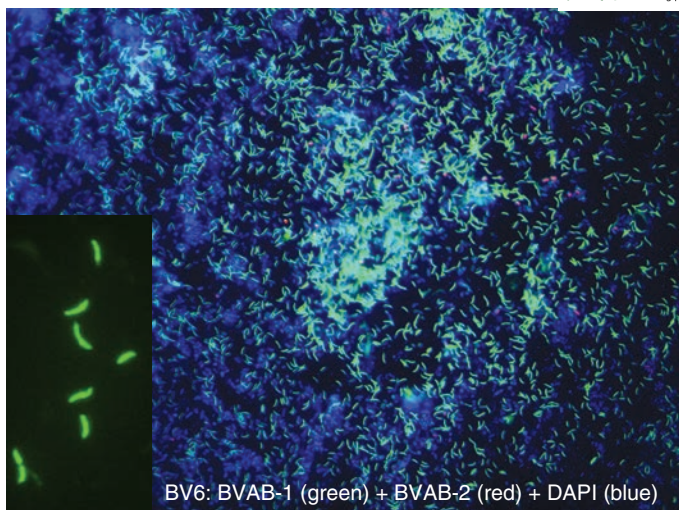


FIGURE 136-2 Broad-range polymerase chain reaction amplification of 16S rDNA in vaginal fluid from a woman with bacterial vaginosis (BV) shows a field of bacteria hybridizing with probes for BV-associated bacterium 1 (BVAB-1, visible as a thin, curved green rod) and for BVAB-2 (red). The inset shows that BVAB-1 has a morphology similar to that of *Mobiluncus* (curved rod). (Adapted from DN Fredricks et al: Molecular identification of bacteria associated with bacterial vaginosis. *N Engl J Med* 353:1899, 2005.)

coated with coccobacillary organisms, which have a granular appearance and indistinct borders; Fig. 136-3) on a wet mount prepared by mixing vaginal secretions with normal saline in a ratio of ~1:1.

TREATMENT

Bacterial Vaginosis

The standard dosage of oral metronidazole for the treatment of BV is 500 mg twice daily for 7 days. Intravaginal treatment with 2% clindamycin cream (one full applicator [5 g containing 100 mg of clindamycin phosphate] each night for 7 nights) or with 0.75% metronidazole gel (one full applicator [5 g containing 37.5 mg of metronidazole] twice daily for 5 days) is also approved for use in the United States and does not elicit systemic adverse reactions; the response to both of these treatments is similar to the response to oral metronidazole. Another nitroimidazole given orally, secnidazole, is also effective (single 2-g dose). Other alternatives include oral clindamycin (300 mg twice daily for 7 days), clindamycin

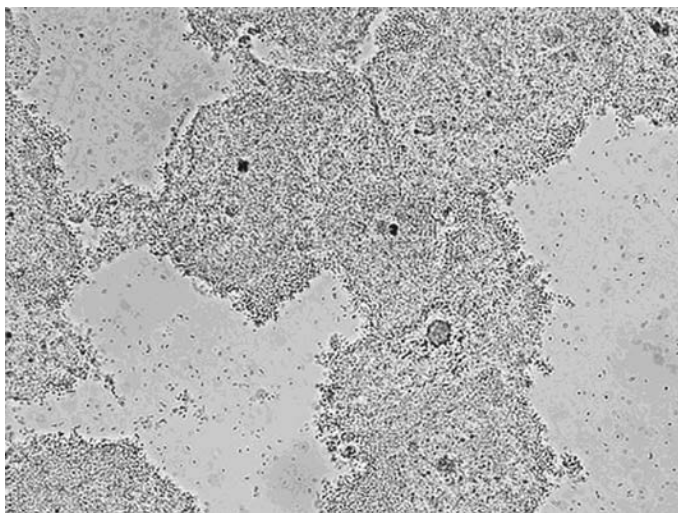


FIGURE 136-3 Wet mount of vaginal fluid showing typical clue cells from a woman with bacterial vaginosis. Note the obscured epithelial cell margins and the granular appearance attributable to many adherent bacteria (×400). (Photograph provided by Lorna K. Rabe, reprinted with permission from S Hillier et al, in KK Holmes et al [eds]: *Sexually Transmitted Diseases*, 4th ed. New York, McGraw-Hill, 2008.)

ovules (100 g intravaginally once at bedtime for 3 days), and oral tinidazole (1 g daily for 5 days or 2 g daily for 3 days). Unfortunately, recurrence over the long term (i.e., several months later) is distressingly common after either oral or intravaginal treatment. A randomized trial comparing intravaginal gel containing 37.5 mg of metronidazole with a suppository containing 500 mg of metronidazole plus nystatin (the latter not marketed in the United States) showed significantly higher rates of recurrence with the 37.5-mg regimen; this result suggests that higher metronidazole dosages may be important in topical intravaginal therapy. Recurrences can be significantly lessened with the twice-weekly use of suppressive intravaginal metronidazole gel. The goal of replenishing the vaginal lactobacilli that sustain vaginal health has recently been bolstered by a randomized trial that demonstrated that weekly vaginal administration of *Lactobacillus crispatus* CTV-05 (LACTIN-V) reduced rates of recurrent BV by approximately one-third.

A meta-analysis of 18 studies concluded that BV during pregnancy substantially increased the risk of preterm delivery and of spontaneous abortion. However, in most studies, topical intravaginal treatment of BV with clindamycin during pregnancy has not reduced adverse pregnancy outcomes. Numerous trials of oral metronidazole treatment during pregnancy have given inconsistent results, and recent reviews have concluded that antenatal treatment of women with BV—including those with previous preterm delivery—did not reduce the risk of preterm delivery. The U.S. Preventive Services Task Force thus recommends against routine screening of pregnant women for BV.

Vulvovaginal Pruritus, Burning, or Irritation Vulvovaginal candidiasis produces vulvar pruritus, burning, or irritation, generally without symptoms of increased vaginal discharge or malodor. Genital herpes can produce similar symptoms, with lesions sometimes difficult to distinguish from the fissures and inflammation caused by candidiasis. Signs of vulvovaginal candidiasis include vulvar erythema, edema, fissures, and tenderness. With candidiasis, a white scanty vaginal discharge sometimes takes the form of white thrush-like plaques or cottage cheese-like curds adhering loosely to the vaginal epithelium. *C. albicans* accounts for nearly all cases of symptomatic vulvovaginal candidiasis, which probably arise from endogenous strains of *C. albicans* that have colonized the vagina or the intestinal tract. Complicated vulvovaginal candidiasis includes cases that recur four or more times per year; are unusually severe; are caused by non-*albicans* *Candida* species; or occur in women with uncontrolled diabetes, debilitation, immunosuppression, or pregnancy.

In addition to compatible clinical symptoms, the diagnosis of vulvovaginal candidiasis involves the demonstration of pseudohyphae or hyphae by microscopic examination of vaginal fluid mixed with saline or 10% KOH or subjected to Gram's staining. Microscopic examination is less sensitive than culture but correlates better with symptoms. Culture is typically reserved for cases that do not respond to standard first-line antimycotic agents and is undertaken to rule out imidazole or azole resistance (often associated with *Candida glabrata*) or before the initiation of suppressive antifungal therapy for recurrent disease.

TREATMENT

Vulvovaginal Pruritus, Burning, or Irritation

Symptoms and signs of vulvovaginal candidiasis warrant treatment, usually intravaginal administration of any of several imidazole antibiotics (e.g., miconazole or clotrimazole) for 3–7 days or of a single dose of oral fluconazole (Table 136-5). Over-the-counter marketing of such preparations has reduced the cost of care and made treatment more convenient for many women with recurrent yeast vulvovaginitis. However, most women who purchase these preparations do not have vulvovaginal candidiasis, whereas many have other vaginal infections that require different treatment. Therefore, only women with classic symptoms of vulvar pruritus and a history of previous episodes of yeast vulvovaginitis documented

by an experienced clinician should self-treat. Short-course topical intravaginal azole drugs are effective for the treatment of uncomplicated vulvovaginal candidiasis (e.g., clotrimazole, two 100-mg vaginal tablets daily for 3 days; or miconazole, a 1200-mg vaginal suppository as a single dose). Single-dose oral treatment with fluconazole (150 mg) is also effective and is preferred by many patients. Management of complicated cases (see above) and those that do not respond to the usual intravaginal or single-dose oral therapy often involves prolonged or periodic oral therapy; this situation is discussed extensively in the 2015 CDC STD treatment guidelines (<http://www.cdc.gov/std/treatment>). Treatment of sexual partners is not routinely indicated.

Other Causes of Vaginal Discharge or Vaginitis In the ulcerative vaginitis associated with staphylococcal toxic shock syndrome, *Staphylococcus aureus* should be promptly identified in vaginal fluid by Gram's stain and by culture. In desquamative inflammatory vaginitis, smears of vaginal fluid reveal neutrophils, massive vaginal epithelial cell exfoliation with increased numbers of parabasal cells, and gram-positive cocci; this syndrome may respond to treatment with 2% clindamycin cream, often given in combination with topical steroid preparations for several weeks. Additional causes of vaginitis and vulvovaginal symptoms include retained foreign bodies (e.g., tampons), cervical caps, vaginal spermicides, vaginal antiseptic preparations or douches, vaginal epithelial atrophy (in postmenopausal women or during prolonged breast-feeding in the postpartum period), allergic reactions to latex condoms, vaginal aphthae associated with HIV infection or Behçet's syndrome, and vestibulitis.

■ MUCOPURULENT CERVICITIS

Mucopurulent cervicitis (MPC) refers to inflammation of the columnar epithelium and subepithelium of the endocervix and of any contiguous columnar epithelium that lies exposed in an ectopic position on the ectocervix. MPC in women represents the "silent partner" of urethritis in men, being equally common and often caused by the same agents (*N. gonorrhoeae*, *C. trachomatis*, *M. genitalium*); however, MPC is more difficult than urethritis to recognize, given the nonspecific nature of symptoms (e.g., abnormal vaginal discharge) and the need for visualization by pelvic examination. As the most common manifestation of these serious bacterial infections in women, MPC can be a harbinger or sign of upper genital tract infection, also known as *pelvic inflammatory disease* (PID; see below). In pregnant women, MPC can lead to obstetric complications. In the pre-NAAT era, more than one-third of cervicovaginal specimens tested for *C. trachomatis*, *N. gonorrhoeae*, *M. genitalium*, HSV, and *T. vaginalis* revealed no identifiable etiology for MPC (Fig. 136-4). More recent studies employing NAATs for these pathogens have still failed to identify a microbiologic etiology in nearly one-half of women with MPC. Individual bacteria associated with BV may also elicit an inflammatory reaction at the cervix; thus, BV may be a cause of MPC.

The diagnosis of MPC rests on the detection of cardinal signs at the cervix, including yellow mucopurulent discharge from the cervical os, endocervical bleeding upon gentle swabbing, and edematous cervical ectopy (see below); the latter two findings are somewhat more common with MPC due to chlamydial infection, but signs alone do not allow a distinction among the causative pathogens. Unlike the endocervicitis produced by gonococcal or chlamydial infection, cervicitis caused by HSV produces ulcerative lesions on the stratified squamous epithelium of the ectocervix as well as on the columnar epithelium. Yellow cervical mucus on a white swab removed from the endocervix indicates the presence of polymorphonuclear leukocytes (PMNs). Gram's staining may confirm their presence, although it adds relatively little to the diagnostic value of assessment for cervical signs. The presence of ≥ 20 PMNs per 1000 $\times$ microscopic field within strands of cervical mucus not contaminated by vaginal squamous epithelial cells or vaginal bacteria indicates endocervicitis. Detection of intracellular gram-negative diplococci in carefully collected endocervical mucus is quite specific but $\leq 50\%$ sensitive for gonorrhea. Therefore, NAATs for *N. gonorrhoeae* and *C. trachomatis* are always indicated in the evaluation of MPC, as

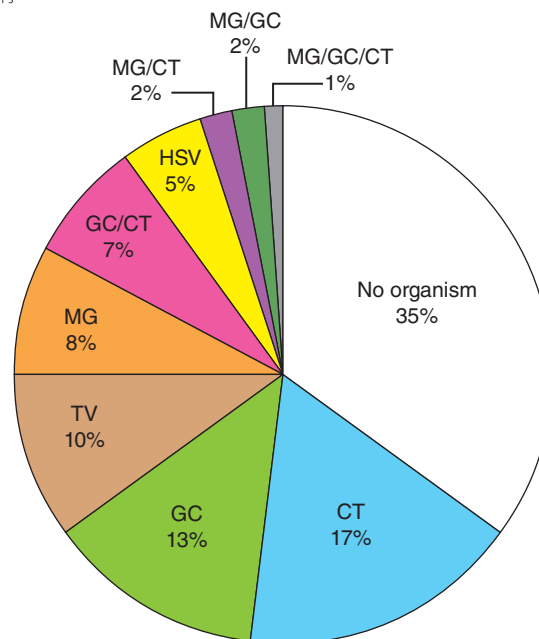


FIGURE 136-4 Organisms detected among female sexually transmitted disease clinic patients with mucopurulent cervicitis (n = 167). CT, *Chlamydia trachomatis*; GC, gonococcus; HSV, herpes simplex virus; MG, *Mycoplasma genitalium*; TV, *Trichomonas vaginalis*. (Courtesy of Dr. Lisa Manhart; with permission.)

is a careful evaluation of vaginal discharge for the causes of vaginitis discussed above.

TREATMENT

Mucopurulent Cervicitis

Although the above criteria for MPC are neither highly specific nor highly predictive of gonococcal or chlamydial infection in some settings, the 2015 CDC STD guidelines call for consideration of empirical treatment for MPC, pending test results, in most cases. Presumptive treatment with antibiotics active against *C. trachomatis* should be provided for women at increased risk for this common STI (risk factors: age < 25 years, new or multiple sex partners, and unprotected sex), especially if follow-up cannot be ensured. Concurrent therapy for gonorrhea is indicated if the prevalence of this infection is substantial in the relevant patient population (e.g., young adults, a clinic with documented high prevalence). In this situation, therapy should include a single-dose regimen effective for gonorrhea plus treatment for chlamydial infection, as outlined in Table 136-4 for the treatment of urethritis. In settings where gonorrhea is much less common than chlamydial infection, initial therapy for chlamydial infection alone suffices, pending test results for gonorrhea. The etiology and potential benefit of treatment for endocervicitis not associated with gonorrhea or chlamydial infection have not been established. Although the antimicrobial susceptibility of *M. genitalium* is not yet well defined, the organism frequently persists after doxycycline therapy, and it currently seems reasonable to use azithromycin to treat possible *M. genitalium* infection in such cases. With resistance of *M. genitalium* to azithromycin now recognized, moxifloxacin may be a reasonable alternative. The sexual partner(s) of a woman with MPC should be examined and given a regimen similar to that chosen for the woman unless results of tests for gonorrhea or chlamydial infection in either partner warrant different therapy or no therapy.

■ CERVICAL ECTOPY

Cervical ectopy, often mislabeled "cervical erosion," is easily confused with infectious endocervicitis. Ectopy represents the presence of the one-cell-thick columnar epithelium extending from the endocervix

out onto the visible ectocervix. In ectopy, the cervical os may contain clear or slightly cloudy mucus but usually not yellow mucopus. Colposcopy shows intact epithelium. Normally found during adolescence and early adulthood, ectopy gradually recedes through the second and third decades of life, as squamous metaplasia replaces the ectopic columnar epithelium. Oral contraceptive use favors the persistence or reappearance of ectopy, while smoking apparently accelerates squamous metaplasia. Cauterization of ectopy is not warranted. Ectopy may render the cervix more susceptible to infection with *N. gonorrhoeae*, *C. trachomatis*, or HIV.

■ PELVIC INFLAMMATORY DISEASE

The term *pelvic inflammatory disease* (PID) usually refers to infection that ascends from the cervix or vagina to involve the endometrium and/or fallopian tubes. Infection can extend beyond the reproductive tract to cause pelvic peritonitis, generalized peritonitis, perihepatitis, perisplenitis, or pelvic abscess. Rarely, infection not related to specific sexually transmitted pathogens extends secondarily to the pelvic organs (1) from adjacent foci of inflammation (e.g., appendicitis, regional ileitis, or diverticulitis) or BV, (2) as a result of hematogenous dissemination (e.g., of tuberculosis or staphylococcal bacteremia), or (3) as a complication of certain tropical diseases (e.g., schistosomiasis). Intrauterine infection can be primary (spontaneously occurring and usually sexually transmitted) or secondary to invasive intrauterine surgical procedures (e.g., dilation and curettage, termination of pregnancy, insertion of an intrauterine device [IUD], or hysterosalpingography) or to parturition.

Etiology The agents most often implicated in acute PID include the primary causes of endocervicitis (*N. gonorrhoeae*, *C. trachomatis*, and *M. genitalium*) and anaerobes associated with BV. In general, PID is most often caused by *N. gonorrhoeae* in settings where there is a high incidence of gonorrhea. *M. genitalium* has also been significantly associated with histopathologic diagnoses of endometritis and with salpingitis.

Anaerobic and facultative organisms (especially *Prevotella* species, peptostreptococci, *E. coli*, *Haemophilus influenzae*, and group B streptococci) as well as genital mycoplasmas have been isolated from the peritoneal fluid or fallopian tubes in a varying proportion (typically one-fourth to one-third) of women with PID studied in the United States. The difficulty of determining the exact microbial etiology of an individual case of PID—short of using invasive procedures for specimen collection—has implications for the approach to empirical antimicrobial treatment of this infection.

Epidemiology In the United States, the estimated annual number of initial visits to physicians' offices for PID by women 15–44 years of age fell from an average of 400,000 during the 1980s to 250,000 in 1999 and then to 51,000 in 2014. Hospitalizations for acute PID in the United States also declined steadily throughout the 1980s and early 1990s but have remained fairly constant at 70,000–100,000 per year since 1995. Important risk factors for acute PID include the presence of endocervical infection or BV, a history of salpingitis or of recent vaginal douching, and recent insertion of an IUD. Certain other iatrogenic factors, such as dilation and curettage or cesarean section, can increase the risk of PID, especially among women with endocervical gonococcal or chlamydial infection or BV. Symptoms of *N. gonorrhoeae*-associated and *C. trachomatis*-associated PID often begin during or soon after the menstrual period; this timing suggests that menstruation is a risk factor for ascending infection from the cervix and vagina. Experimental inoculation of the fallopian tubes of nonhuman primates has shown that repeated exposure to *C. trachomatis* leads to the greatest degree of tissue inflammation and damage; thus, immunopathology probably contributes to the pathogenesis of chlamydial salpingitis. Women using oral contraceptives appear to be at decreased risk of symptomatic PID, and tubal sterilization reduces the risk of salpingitis by preventing intraluminal spread of infection into the tubes.

Clinical Manifestations • **ENDOMETRITIS: A CLINICAL PATHOLOGIC SYNDROME** A study of women with clinically suspected PID who were undergoing both endometrial biopsy and laparoscopy

showed that those with endometritis alone differed from those who also had salpingitis in significantly less often having lower quadrant, adnexal, or cervical motion or abdominal rebound tenderness; fever; or elevated C-reactive protein levels. In addition, women with endometritis alone differed from those with neither endometritis nor salpingitis in more often having gonorrhea, chlamydial infection, and risk factors such as douching or IUD use. Thus, women with endometritis alone were intermediate between those with neither endometritis nor salpingitis and those with salpingitis with respect to risk factors, clinical manifestations, cervical infection prevalence, and elevated C-reactive protein level. Women with endometritis alone are at lower risk of subsequent tubal occlusion and resulting infertility than are those with salpingitis.

SALPINGITIS Symptoms of nontuberculous salpingitis classically evolve from a yellow or malodorous vaginal discharge caused by MPC and/or BV to midline abdominal pain and abnormal vaginal bleeding caused by endometritis and then to bilateral lower abdominal and pelvic pain caused by salpingitis, with nausea, vomiting, and increased abdominal tenderness if peritonitis develops.

The abdominal pain in nontuberculous salpingitis is usually described as dull or aching. In some cases, pain is lacking or atypical, but active inflammatory changes are found in the course of an unrelated evaluation or procedure, such as a laparoscopic evaluation for infertility. Abnormal uterine bleeding precedes or coincides with the onset of pain in ~40% of women with PID, symptoms of urethritis (dysuria) occur in 20%, and symptoms of proctitis (anorectal pain, tenesmus, and rectal discharge or bleeding) are occasionally seen in women with gonococcal or chlamydial infection.

Speculum examination shows evidence of MPC (yellow endocervical discharge, easily induced endocervical bleeding) in the majority of women with gonococcal or chlamydial PID. Cervical motion tenderness is produced by stretching of the adnexal attachments on the side toward which the cervix is pushed. Bimanual examination reveals uterine fundal tenderness due to endometritis and abnormal adnexal tenderness due to salpingitis that is usually, but not necessarily, bilateral. Adnexal swelling is palpable in about one-half of women with acute salpingitis, but evaluation of the adnexae in a patient with marked tenderness is not reliable. The initial temperature is >38°C in only about one-third of patients with acute salpingitis. Laboratory findings include elevation of the erythrocyte sedimentation rate (ESR) in 75% of patients with acute salpingitis and elevation of the peripheral white blood cell count in up to 60%.

Unlike nontuberculous salpingitis, genital tuberculosis often occurs in older women, many of whom are postmenopausal. Presenting symptoms include abnormal vaginal bleeding, pain (including dysmenorrhea), and infertility. About one-quarter of these women have had adnexal masses. Endometrial biopsy shows tuberculous granulomas and provides optimal specimens for culture.

PERIHEPATITIS AND PERIAPPENDICITIS Pleuritic upper abdominal pain and tenderness, usually localized to the right upper quadrant (RUQ), develop in 3–10% of women with acute PID. Symptoms of perihepatitis arise during or after the onset of symptoms of PID and may overshadow lower abdominal symptoms, thereby leading to a mistaken diagnosis of cholecystitis. In perhaps 5% of cases of acute salpingitis, early laparoscopy reveals perihepatic inflammation ranging from edema and erythema of the liver capsule to exudate with fibrinous adhesions between the visceral and parietal peritoneum. When treatment is delayed and laparoscopy is performed late, dense “violin-string” adhesions can be seen over the liver; chronic exertional or positional RUQ pain ensues when traction is placed on the adhesions. Although perihepatitis, also known as the *Fitz-Hugh-Curtis syndrome*, was for many years specifically attributed to gonococcal salpingitis, most cases are now attributed to chlamydial salpingitis. In patients with chlamydial salpingitis, serum titers of microimmunofluorescent antibody to *C. trachomatis* are typically much higher when perihepatitis is present than when it is absent.

Physical findings include RUQ tenderness and usually include adnexal tenderness and cervicitis, even in patients whose symptoms do not suggest salpingitis. Results of liver function tests and RUQ

1088 ultrasonography are nearly always normal. The presence of MPC and pelvic tenderness in a young woman with subacute pleuritic RUQ pain and normal ultrasonography of the gallbladder points to a diagnosis of perihepatitis.

Periappendicitis (appendiceal serositis without involvement of the intestinal mucosa) has been found in ~5% of patients undergoing appendectomy for suspected appendicitis and can occur as a complication of gonococcal or chlamydial salpingitis.

Among women with salpingitis, HIV infection is associated with increased severity of salpingitis and with tuboovarian abscess requiring hospitalization and surgical drainage. Nonetheless, among women with HIV infection and salpingitis, the clinical response to conventional antimicrobial therapy (coupled with drainage of tuboovarian abscess, when found) has usually been satisfactory.

Diagnosis Treatment appropriate for PID must not be withheld from patients who have an equivocal diagnosis; it is better to err on the side of overdiagnosis and overtreatment. On the other hand, it is essential to differentiate between salpingitis and other pelvic pathology, particularly surgical emergencies such as appendicitis and ectopic pregnancy or the chronic syndrome of endometriosis.

Nothing short of laparoscopy definitively identifies salpingitis, but routine laparoscopy to confirm suspected salpingitis is generally impractical. Most patients with acute PID have lower abdominal pain of <3 weeks' duration, pelvic tenderness on bimanual pelvic examination, and evidence of lower genital tract infection (e.g., MPC). Approximately 60% of such patients have salpingitis at laparoscopy, and perhaps 10–20% have endometritis alone. Among the patients with these findings, a rectal temperature >38°C, a palpable adnexal mass, and elevation of the ESR to >15 mm/h also raise the probability of salpingitis, which has been found at laparoscopy in 68% of patients with one of these additional findings, 90% of patients with two, and 96% of patients with three. However, only 17% of all patients with laparoscopy-confirmed salpingitis have had all three additional findings.

In a woman with pelvic pain and tenderness, increased numbers of PMNs (30 per 1000× microscopic field in strands of cervical mucus) or leukocytes outnumbering epithelial cells in vaginal fluid (in the absence of trichomonal vaginitis, which also produces PMNs in vaginal discharge) increase the predictive value of a clinical diagnosis of acute PID, as do onset with menses, history of recent abnormal menstrual bleeding, presence of an IUD, history of salpingitis, and sexual exposure to a male with urethritis. Appendicitis or another disorder of the gut is favored by the early onset of anorexia, nausea, or vomiting; the onset of pain later than day 14 of the menstrual cycle; or unilateral pain limited to the right or left lower quadrant. Whenever the diagnosis of PID is being considered, serum assays for human β-chorionic gonadotropin should be performed; these tests are usually positive with ectopic pregnancy. Ultrasonography and MRI can be useful for the identification of tuboovarian or pelvic abscess. MRI of the tubes can also show increased tubal diameter, intratubal fluid, or tubal wall thickening in cases of salpingitis.

The primary value of laparoscopy in women with lower abdominal pain is for the exclusion of other surgical problems that cannot be resolved with noninvasive imaging. Some of the most common or serious problems that may be confused with salpingitis (e.g., acute appendicitis, ectopic pregnancy, corpus luteum bleeding, ovarian tumor) are unilateral. Unilateral pain or pelvic mass, although not incompatible with PID, is a strong indication for laparoscopy unless the clinical picture warrants laparotomy instead. Atypical clinical findings such as the absence of lower genital tract infection, a missed menstrual period, a positive pregnancy test, or failure to respond to appropriate therapy are other common indications for laparoscopy. Endometrial biopsy is relatively sensitive and specific for the diagnosis of endometritis, which correlates well with the presence of salpingitis.

Vaginal or endocervical swab specimens should be obtained for NAATs for *N. gonorrhoeae* and *C. trachomatis*. At a minimum, vaginal fluid should be evaluated for the presence of PMNs, and endocervical secretions ideally should be assessed by Gram's staining for PMNs and gram-negative diplococci, which indicate gonococcal infection. The

clinical diagnosis of PID made by expert gynecologists is confirmed by laparoscopy or endometrial biopsy in ~90% of women who also have cultures positive for *N. gonorrhoeae* or *C. trachomatis*. Even among women with no symptoms suggestive of acute PID who were attending an STD clinic or a gynecology clinic in Pittsburgh, endometritis was significantly associated with endocervical gonorrhea or chlamydial infection or with BV, being detected in 26%, 27%, and 15% of women with these conditions, respectively.

TREATMENT

Pelvic Inflammatory Disease

Recommended combination regimens for ambulatory or parenteral management of PID are presented in Table 136-6. Women managed as outpatients should receive a combined regimen with broad activity, such as ceftriaxone (to cover possible gonococcal infection) followed by doxycycline (to cover possible chlamydial infection). Metronidazole should be strongly considered to enhance activity against anaerobes, especially if BV or trichomoniasis are present or there is a recent (3-week) history of gynecologic instrumentation; in a recent randomized trial, the addition of metronidazole to ceftriaxone and doxycycline effected reduction in endometrial anaerobes, *M. genitalium*, and pelvic tenderness.

The CDC STD treatment guidelines recommend initiation of empirical treatment for PID in sexually active young women and other women at risk for PID if they are experiencing pelvic or lower abdominal pain, if no other cause for the pain can be identified, and if pelvic examination reveals one or more of the following criteria for PID: cervical motion tenderness, uterine tenderness, or adnexal tenderness. Women with suspected PID can be treated as either outpatients or inpatients. In the multicenter Pelvic Inflammatory Disease Evaluation and Clinical Health (PEACH) trial, 831 women with mild to moderately severe symptoms and signs of PID were randomized to receive either inpatient treatment with IV cefoxitin and doxycycline or outpatient treatment with a single IM dose of cefoxitin plus oral doxycycline. Short-term clinical and microbiologic outcomes and long-term outcomes were equivalent in the two groups. Nonetheless, hospitalization should be considered when (1) the diagnosis is uncertain and surgical emergencies such as appendicitis and ectopic pregnancy cannot be excluded, (2) the patient is pregnant, (3) pelvic abscess is suspected, (4) severe illness or nausea and vomiting preclude outpatient management, (5) the patient has HIV infection, (6) the patient is assessed as unable to follow or tolerate an outpatient regimen, or (7) the patient has failed to respond

TABLE 136-6 Combination Antimicrobial Regimens Recommended for Outpatient Treatment or for Parenteral Treatment of Pelvic Inflammatory Disease

OUTPATIENT REGIMENS ^a	PARENTERAL REGIMENS
Ceftriaxone (500 mg IM once) plus Doxycycline (100 mg PO bid for 14 days) plus^b Metronidazole (500 mg PO bid for 14 days)	Initiate parenteral therapy with either of the following regimens; continue parenteral therapy until 48 h after clinical improvement; then change to outpatient therapy, as described in the text Regimen A Cefotetan (2 g IV q12h) or cefoxitin (2 g IV q6h) plus Doxycycline (100 mg IV or PO q12h) Regimen B Clindamycin (900 mg IV q8h) plus Gentamicin (loading dose of 2 mg/kg IV or IM, then maintenance dose of 1.5 mg/kg q8h)

^aSee text for discussion of options in the patient who is intolerant of cephalosporins.

^bThe addition of metronidazole is recommended particularly if bacterial vaginosis or trichomoniasis is present.

Source: Adapted from Centers for Disease Control and Prevention: MMWR Recomm Rep 64(RR-03):1, 2015.

to outpatient therapy. Some experts also prefer to hospitalize adolescents with PID for initial therapy, although younger women do as well as older women on outpatient therapy.

Currently, no agents other than parenteral cephalosporins provide reliable coverage for gonococcal infection. Thus, adequate oral treatment of women with serious intolerance to cephalosporins is a challenge. If penicillins are an option, amoxicillin/clavulanic acid combined with doxycycline has elicited a short-term clinical response in one trial. Clinical trials performed outside the United States support the effectiveness of oral moxifloxacin. In this case, it is imperative to perform a sensitive diagnostic test for gonorrhea (ideally, a culture to test for antimicrobial susceptibility) before initiation of therapy. For women whose PID involves quinolone-resistant *N. gonorrhoeae*, treatment is uncertain but could include parenteral gentamicin or oral azithromycin, although the latter agent has not been studied for this purpose.

For hospitalized patients, the following two parenteral regimens (Table 136-6) have given nearly identical results in a multicenter randomized trial:

1. Doxycycline plus either cefotetan or cefoxitin: Administration of these drugs should be continued by the IV route for at least 48 h after the patient's condition improves and then followed with oral doxycycline (100 mg twice daily) to complete 14 days of therapy.
2. Clindamycin plus gentamicin in patients with normal renal function: Once-daily administration of gentamicin (with combination of the total daily dose into a single daily dose) has not been evaluated in PID but has been efficacious in other serious infections and could be substituted. Treatment with these drugs should be continued for at least 48 h after the patient's condition improves and then followed with oral doxycycline (100 mg twice daily) or clindamycin (450 mg four times daily) to complete 14 days of therapy. In cases with tuboovarian abscess, clindamycin rather than doxycycline for continued therapy provides better coverage for anaerobic infection.

FOLLOW-UP

Hospitalized patients should show substantial clinical improvement within 3–5 days. Women treated as outpatients should be clinically reevaluated within 72 h. A follow-up telephone survey of women seen in an emergency department and given a prescription for 10 days of oral doxycycline for PID found that 28% never filled the prescription and 41% stopped taking the medication early (after an average of 4.1 days), often because of persistent symptoms, lack of symptoms, or side effects. Women not responding favorably to ambulatory therapy should be hospitalized for parenteral therapy and further diagnostic evaluations, including a consideration of laparoscopy. Male sex partners should be evaluated and treated empirically for gonorrhea and chlamydial infection. After completion of treatment, tests for persistent or recurrent infection with *N. gonorrhoeae* or *C. trachomatis* should be performed if symptoms persist or recur or if the patient has not complied with therapy or has been reexposed to an untreated sex partner.

SURGERY

Surgery is necessary for the treatment of salpingitis only in the face of life-threatening infection (such as rupture or threatened rupture of a tuboovarian abscess) or for drainage of an abscess. Conservative surgical procedures are usually sufficient. Pelvic abscesses can often be drained by posterior colpotomy, and peritoneal lavage can be used for generalized peritonitis.

Prognosis Late sequelae include infertility due to bilateral tubal occlusion, ectopic pregnancy due to tubal scarring without occlusion, chronic pelvic pain, and recurrent salpingitis. The overall post-salpingitis risk of infertility due to tubal occlusion in a large study in Sweden was 11% after one episode of salpingitis, 23% after two episodes, and 54% after three or more episodes. A University of Washington study found a sevenfold increase in the risk of ectopic pregnancy and an eightfold increase in the rate of hysterectomy after PID.

Prevention A randomized controlled trial designed to determine whether selective screening for chlamydial infection reduces the risk of subsequent PID showed that women randomized to undergo screening had a 56% lower rate of PID over the following year than did women receiving the usual care without screening. This report helped prompt U.S. national guidelines for risk-based chlamydial screening of young women to reduce the incidence of PID and the prevalence of post-PID sequelae, while also reducing sexual transmission of *C. trachomatis*. The CDC and the U.S. Preventive Services Task Force recommend that sexually active women ≤ 25 years of age be screened annually for genital chlamydial infection. Despite this recommendation, screening coverage in many primary care settings remains low.

■ ULCERATIVE GENITAL OR PERIANAL LESIONS

Genital ulceration reflects a set of important STIs, most of which sharply increase the risk of sexual acquisition and shedding of HIV. In a 1996 study of genital ulcers in 10 of the U.S. cities with the highest rates of primary syphilis, PCR testing of ulcer specimens demonstrated HSV in 62% of patients, *T. pallidum* in 13%, and *Haemophilus ducreyi* (the cause of chancroid) in 12–20%. Today, genital herpes represents an even higher proportion of genital ulcers in the United States and other industrialized countries.

In Asia and Africa, chancroid (Fig. 136-5) was once considered the most common type of genital ulcer, followed in frequency by primary syphilis and then genital herpes (Fig. 136-6). With increased efforts to control chancroid and syphilis and widespread use of broad-spectrum antibiotics to treat STI-related syndromes, together with more frequent recurrences or persistence of genital herpes attributable to HIV infection, PCR testing of genital ulcers now clearly implicates genital herpes as by far the most common cause of genital ulceration in most developing countries. LGV due to *C. trachomatis* (Fig. 136-7) and donovanosis (granuloma inguinale, due to *Klebsiella granulomatis*; see Fig. 173-1) continue to cause genital ulceration in some developing countries. LGV virtually disappeared in industrialized countries during the first 20 years of the HIV pandemic, but outbreaks are again occurring in Europe (including the United Kingdom), in North America, and in Australia. In these outbreaks, LGV typically presents as proctitis, with or without anal lesions, in men who report unprotected receptive anal intercourse, very often in association with HIV and/or hepatitis C virus infection; the latter may be an acute infection acquired through the same exposure. Other causes of genital ulcers include (1) candidiasis and traumatized genital warts—both readily recognized; (2) lesions due to genital involvement by more widespread dermatoses;



FIGURE 136-5 Chancroid: multiple, painful, punched-out ulcers with undermined borders on the labia occurring after autoinoculation.



FIGURE 136-6 Genital herpes. A relatively mild, superficial ulcer is typically seen in episodic outbreaks. (Courtesy of Michael Remington, University of Washington Virology Research Clinic.)

(3) cutaneous manifestations of systemic diseases such as genital mucosal ulceration in Stevens-Johnson syndrome or Behçet's disease; (4) superinfections of lesions that may originally have been sexually acquired (for example, methicillin-resistant *S. aureus* complicating a genital ulcer due to HSV-2); and (5) localized drug reactions, such as the ulcers occasionally seen with topical paromomycin cream or boric acid preparations.

Diagnosis Although most genital ulcerations cannot be diagnosed confidently on clinical grounds alone, clinical findings (Table 136-7) and epidemiologic considerations can usually guide initial management (Table 136-8) pending results of specific tests. Clinicians should order a rapid serologic test for syphilis in all cases of genital ulcer and treat presumptively while awaiting serology in a patient at high risk (especially MSM). To evaluate lesions except those highly characteristic of infection with HSV (i.e., those with herpetic vesicles), dark-field microscopy, direct immunofluorescence, and a NAAT for *T. pallidum* can be useful but are rarely available. It is important to note that 30% of syphilitic chancres—the primary ulcer of syphilis—are associated with an initially nonreactive syphilis serology. All patients presenting with genital ulceration should be counseled and tested for HIV infection.

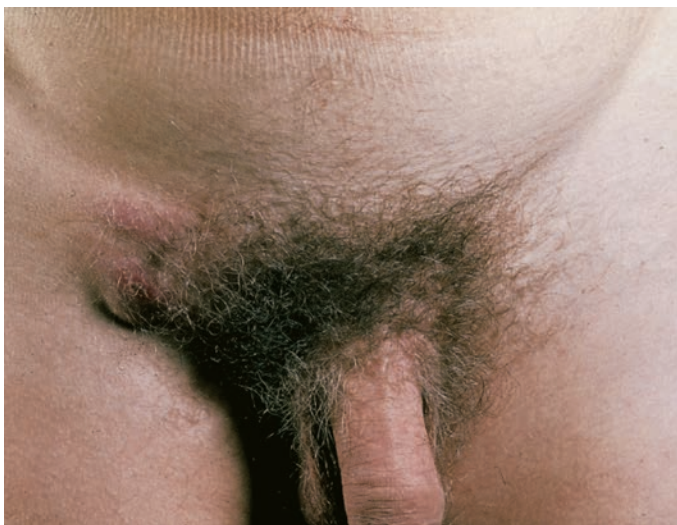


FIGURE 136-7 Lymphogranuloma venereum (LGV): striking tender lymphadenopathy occurring at the femoral and inguinal lymph nodes, separated by a groove made by Poupart's ligament. This "sign-of-the-groove" is not considered specific for LGV; for example, lymphomas may present with this sign.

Typical vesicles or pustules or a cluster of painful ulcers preceded by vesiculopustular lesions suggest genital herpes. These typical clinical manifestations make detection of the virus optional; however, many patients want confirmation of the diagnosis, and differentiation of HSV-1 from HSV-2 has prognostic implications, because the latter causes more frequent genital recurrences and is more infectious to vulnerable sex partners.

Painless, nontender, indurated ulcers with firm, nontender inguinal adenopathy suggest primary syphilis. If results of dark-field examination and a rapid serologic test for syphilis are initially negative, or if these tests are not available, presumptive therapy should be provided on the basis of the individual's risk. With historically high rates of syphilis among MSM in the United States, therapy for this infection should not be withheld pending watchful waiting and/or subsequent detection of seroconversion. Repeated serologic testing for syphilis 1 or 2 weeks after treatment of seronegative primary syphilis usually demonstrates seroconversion.

"Atypical" or clinically trivial ulcers may be more common manifestations of genital herpes than classic vesiculopustular lesions. Specific tests for HSV in such lesions are therefore indicated (Chap. 192). Commercially available type-specific serologic tests for serum antibody to HSV-2 may give negative results, especially when patients present early with the initial episode of genital herpes or when HSV-1 is the cause of genital herpes (as is often the case today). Furthermore, a positive test for antibody to HSV-2 does not prove that the current lesions are herpetic because nearly one-fifth of the general population of the United States (and no doubt a higher proportion of those at risk for other STIs) becomes seropositive for HSV-2 during early adulthood. Although even "type-specific" tests for HSV-2 that are commercially available in the United States are not 100% specific, a positive HSV-2 serology does enable the clinician to tell the patient that he or she has probably had genital herpes, should learn to recognize symptoms, and should avoid sex during recurrences. In addition, because genital shedding and sexual transmission of HSV-2 often occur in the absence of symptoms and signs of recurrent herpetic lesions, persons who have a history of genital herpes or who are seropositive for HSV-2 should consider the use of condoms or suppressive antiviral therapy, both of which can reduce the risk of HSV-2 transmission to a sexual partner.

Demonstration of *H. ducreyi* by culture (or by PCR, where available) is most useful when ulcers are painful and purulent, especially if inguinal lymphadenopathy with fluctuance or overlying erythema is noted; if chancroid is prevalent in the community; or if the patient has recently had a sexual exposure elsewhere in a chancroid-endemic area (e.g., a developing country). Enlarged, fluctuant lymph nodes should be aspirated for culture or PCR to detect *H. ducreyi* as well as for Gram's staining and culture to rule out the presence of other pyogenic bacteria.

When genital ulcers persist beyond the natural history of initial episodes of herpes (2–3 weeks) or of chancroid or syphilis (up to 6 weeks) and do not resolve with syndrome-based antimicrobial therapy, then—in addition to the usual tests for herpes, syphilis, and chancroid—biopsy is indicated to exclude donovanosis as well as carcinoma and other nonvenereal dermatoses.

TREATMENT

Ulcerative Genital or Perianal Lesions

Immediate syndrome-based treatment for acute genital ulcer (after collection of all necessary diagnostic specimens at the first visit) is often appropriate before all test results become available because patients with typical initial or recurrent episodes of genital or anorectal herpes can benefit from prompt oral antiviral therapy (Chap. 192); because early treatment of sexually transmitted causes of genital ulcers decreases further transmission; and because some patients do not return for test results and treatment. A thorough assessment of the patient's sexual-risk profile and medical history is critical in determining the course of initial management. The patient who has risk factors consistent with exposure to syphilis (e.g., a male patient who reports sex with other men or who has

TABLE 136-7 Clinical Features of Genital Ulcers					
FEATURE	SYPHILIS	HERPES	CHANCROID	LYMPHOGRANULOMA VENEREUM	DONOVANOSIS
Incubation period	9–90 days	2–7 days	1–14 days	3 days–6 weeks	1–4 weeks (up to 6 months)
Early primary lesions	Papule	Vesicle	Pustule	Papule, pustule, or vesicle	Papule
Number of lesions	Usually one	Multiple	Usually multiple, may coalesce	Usually one; often not detected, despite lymphadenopathy	Variable
Diameter	5–15 mm	1–2 mm	Variable	2–10 mm	Variable
Edges	Sharply demarcated, elevated, round, or oval	Erythematous	Undermined, ragged, irregular	Elevated, round, or oval	Elevated, irregular
Depth	Superficial or deep	Superficial	Excavated	Superficial or deep	Elevated
Base	Smooth, nonpurulent, relatively nonvascular	Serous, erythematous, nonvascular	Purulent, bleeds easily	Variable, nonvascular	Red and velvety, bleeds readily
Induration	Firm	None	Soft	Occasionally firm	Firm
Pain	Uncommon	Frequently tender	Usually very tender	Variable	Uncommon
Lymphadenopathy	Firm, nontender, bilateral	Firm, tender, often bilateral with initial episode	Tender, may suppurate, loculated, usually unilateral	Tender, may suppurate, loculated, usually unilateral	None; pseudobuboes

Source: Reproduced with permission from RM Ballard, in KK Holmes et al (eds): *Sexually Transmitted Diseases*, 4th ed. New York, McGraw-Hill, 2008.

HIV infection) should generally receive initial treatment for syphilis. Empirical therapy for chancroid should be considered if there has been an exposure in an area of the world where chancroid occurs or if regional lymph node suppuration is evident. Finally, empirical antimicrobial therapy may be indicated if ulcers persist and the diagnosis remains unclear after a week of observation despite attempts to diagnose herpes, syphilis, and chancroid.

■ PROCTITIS, PROCTOCOLITIS, ENTEROCOLITIS, AND ENTERITIS

Sexually acquired *proctitis*, with inflammation limited to the rectal mucosa (the distal 10–12 cm), results from direct rectal inoculation of typical STD pathogens. In contrast, inflammation extending from the rectum to the colon (*proctocolitis*), involving both the small and the

large bowel (*enterocolitis*), or involving the small bowel alone (*enteritis*) can result from ingestion of typical intestinal pathogens through oral-anal exposure during sexual contact. Anorectal pain and mucopurulent, bloody rectal discharge suggest proctitis or proctocolitis. Proctitis commonly produces tenesmus (causing frequent attempts to defecate, but not true diarrhea) and constipation, whereas proctocolitis and enterocolitis more often cause true diarrhea. In all three conditions, anoscopy usually shows mucosal exudate and easily induced mucosal bleeding (i.e., a positive “wipe test”), sometimes with petechiae or mucosal ulcers. Exudate should be sampled for Gram’s staining and other microbiologic studies. Sigmoidoscopy or colonoscopy shows inflammation limited to the rectum in proctitis or disease extending at least up into the sigmoid colon in proctocolitis.

The AIDS era brought an extraordinary shift in the clinical and etiologic spectrum of intestinal infections among MSM. The number of cases of the acute intestinal STIs described above fell as high-risk sexual behaviors became less common in this group. At the same time, the number of AIDS-related opportunistic intestinal infections increased rapidly, many associated with chronic or recurrent symptoms. The incidence of these opportunistic infections has since fallen with increasingly widespread coverage of HIV-infected persons with effective antiretroviral therapy. Two species initially isolated in association with intestinal symptoms in MSM—now known as *Helicobacter cinaedi* and *Helicobacter fennelliae*—have both been isolated from the blood of HIV-infected men and other immunosuppressed persons, often in association with a syndrome of multifocal dermatitis and arthritis.

Acquisition of HSV, *N. gonorrhoeae*, or *C. trachomatis* (including LGV strains of *C. trachomatis*) during receptive anorectal intercourse causes most cases of infectious proctitis in women and MSM. Primary and secondary syphilis can also produce anal or anorectal lesions, with or without symptoms. Gonococcal or chlamydial proctitis typically involves the most distal rectal mucosa and the anal crypts and is clinically mild, without systemic manifestations. In contrast, primary proctitis due to HSV and proctocolitis due to the strains of *C. trachomatis* that cause LGV usually produce severe anorectal pain and often cause fever. Perianal ulcers and inguinal lymphadenopathy, most commonly due to HSV, can also occur with LGV or syphilis. Sacral nerve root radiculopathies, usually presenting as urinary retention, laxity of the anal sphincter, or constipation, may complicate primary herpetic proctitis. In LGV, rectal biopsy typically shows crypt abscesses, granulomas, and giant cells—findings resembling those in Crohn’s disease; such findings should always prompt rectal culture and serology for LGV, which is a curable infection. Syphilis can also produce rectal granulomas, usually in association with infiltration by plasma cells or other mononuclear cells. Syphilis, LGV, and HSV infection involving the rectum can produce perirectal adenopathy that is sometimes mistaken

TABLE 136-8 Initial Management of Genital or Perianal Ulcer

Causative Pathogens
HSV <i>Treponema pallidum</i> (primary syphilis) <i>Haemophilus ducreyi</i> (chancroid)
Usual Initial Laboratory Evaluation
Dark-field examination (if available), direct FA, or PCR for <i>T. pallidum</i> RPR, VDRL, or EIA serologic test for syphilis ^a Culture, direct FA, ELISA, or PCR for HSV HSV-2-specific serology (consider) In chancroid-endemic area: PCR or culture for <i>H. ducreyi</i>
Initial Treatment
Herpes confirmed or suspected (history or sign of vesicles): Treat for genital herpes with acyclovir, valacyclovir, or famciclovir.
Syphilis confirmed (dark-field, FA, or PCR showing <i>T. pallidum</i> , or RPR reactive): Benzathine penicillin (2.4 million units IM once to patient, to recent [e.g., within 3 months] seronegative partner[s], and to all seropositive partners) ^b
Chancroid confirmed or suspected (diagnostic test positive, or HSV and syphilis excluded, and persistent lesion): Ciprofloxacin (500 mg PO as single dose) or Ceftriaxone (250 mg IM as single dose) or Azithromycin (1 g PO as single dose)

^aIf results are negative but primary syphilis is suspected, treat presumptively when indicated by epidemiologic and sexual risk assessment; repeat in 1 week. ^bThe same treatment regimen is also effective in HIV-infected persons with early syphilis.

Abbreviations: EIA, enzyme immunoassay; ELISA, enzyme-linked immunosorbent assay; FA, fluorescent antibody; HSV, herpes simplex virus; PCR, polymerase chain reaction; RPR, rapid plasma reagin; VDRL, Venereal Disease Research Laboratory.

1092 for malignancy; syphilis, LGV, HSV infection, and chancroid involving the anus can produce inguinal adenopathy because anal lymphatics drain to inguinal lymph nodes.

Diarrhea and abdominal bloating or cramping pain without anorectal symptoms and with normal findings on anoscopy and sigmoidoscopy occur with inflammation of the small intestine (enteritis) or with proximal colitis. In MSM without HIV infection, enteritis is often attributable to *Giardia lamblia*. Sexually acquired proctocolitis is most often due to *Campylobacter* or *Shigella* species.

TREATMENT

Proctitis, Proctocolitis, Enterocolitis, and Enteritis

Acute proctitis in persons who have practiced receptive anorectal intercourse is usually sexually acquired. Such patients should undergo anoscopy to detect rectal ulcers or vesicles and petechiae after swabbing of the rectal mucosa; to examine rectal exudates for PMNs and gram-negative diplococci; and to obtain rectal swab specimens for testing for rectal gonorrhea, chlamydial infection, herpes, and syphilis. Pending test results, patients with proctitis should receive empirical syndromic treatment—e.g., with ceftriaxone (a single IM dose of 500 mg for gonorrhea) plus doxycycline (100 mg by mouth twice daily for 7 days for possible chlamydial infection) plus treatment for herpes or syphilis if indicated. If LGV proctitis is proven or suspected, the recommended treatment is doxycycline (100 mg by mouth twice daily for 21 days); alternatively, 1 g of azithromycin once a week for 3 weeks is likely to be effective but is little studied.

PREVENTION AND CONTROL OF STIs

Prevention and control of STIs require the following:

1. Reduction of the average rate of sexual exposure to STIs through alteration of sexual risk behaviors and behavioral norms among both susceptible and infected persons in all population groups. The necessary changes include reduction in the total number of sexual partners and the number of concurrent sexual partners. The U.S. Preventive Services Task Force recommends intensive behavioral counseling for all sexually active adolescents and adults who are at increased risk for STIs (grade B recommendation). Motivational interviewing is one approach that has elicited behavioral changes, including safer sex practices and more consistent contraception, that contribute to these goals.
2. Reduction of the efficiency of transmission through the promotion of safer sexual practices, the use of condoms during casual or commercial sex, vaccination against HBV and HPV infection, male circumcision (which reduces risk of acquisition of HIV infection, chancroid, and perhaps other STIs), and a growing number of other approaches (e.g., early detection and treatment of other STIs to reduce the efficiency of sexual transmission of HIV). Longitudinal studies have shown that consistent condom use is associated with significant protection of both males and females against all STIs that have been examined, including HIV, HPV, and HSV infections as well as gonorrhea and chlamydial infection. The only exceptions are probably sexually transmitted *Pthirus pubis* and *Sarcoptes scabiei* infestations.
3. Shortening of the duration of infectivity of STIs through early detection and curative or suppressive treatment of patients and their sexual partners. The availability of curative therapy for hepatitis C virus infection and suppressive therapy for HBV infection exemplifies new opportunities for shortening infectivity in major STIs.

Financial and time constraints imposed by many clinical practices, along with the reluctance of some clinicians to ask questions about stigmatized sexual behaviors, often curtail screening and prevention services. As outlined in Fig. 136-8, the success of clinicians' efforts to detect and treat STIs depends in part on societal efforts to teach young people how to recognize symptoms of STIs; to motivate individuals

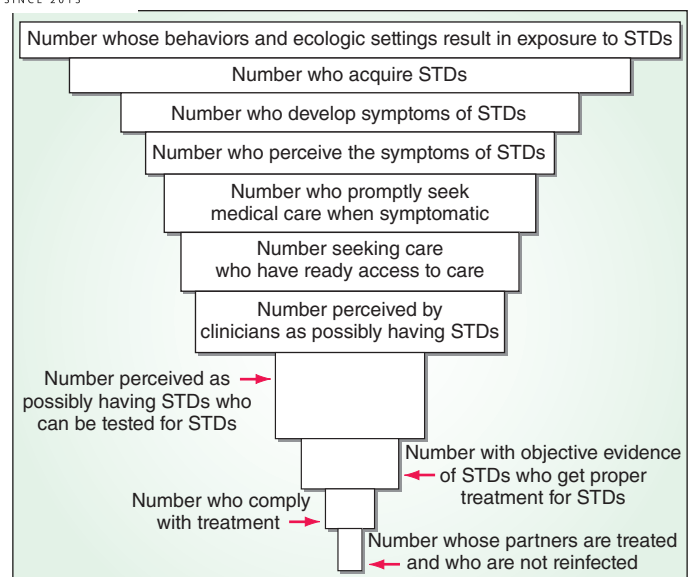


FIGURE 136-8 Critical control points for preventive and clinical interventions against sexually transmitted diseases (STDs). (Adapted from HT Waller and MA Piot: *Bull World Health Organ* 41:75, 1969 and 43:1, 1970; and from "Resource allocation model for public health planning—a case study of tuberculosis control," *Bull World Health Organ* 48[Suppl], 1973.)

with symptoms to seek care promptly; to educate persons who are at risk but have no symptoms about what tests they should undergo routinely; and to make high-quality, appropriate care accessible, affordable, and acceptable, especially to the young indigent patients most likely to acquire an STI.

STI RISK ASSESSMENT

Because many infected individuals develop no symptoms or fail to recognize and report symptoms, clinicians should routinely perform an STI risk assessment for teenagers and young adults as a guide to selective screening. As stated earlier, the U.S. Preventive Services Task Force recommends screening sexually active female patients ≤ 25 years of age for *C. trachomatis* whenever they present for health care (at least once a year); older women should be tested if they have more than one sexual partner, have begun a new sexual relationship since the previous test, or have another STI diagnosed. In women 25–29 years of age, chlamydial infection is uncommon but still may reach a prevalence of 3–5% in some settings; information provided by women in this age group on a sex partner's concurrency (whether a male partner has had another sex partner during the time they have been together) is helpful in identifying women at increased risk. In some regions of the United States, widespread selective screening and treatment of young women for cervical *C. trachomatis* infection have been associated with a 50–60% drop in prevalence. Such screening and treatment also protect the individual woman from PID. Sensitive urine-based genetic amplification tests permit expansion of screening to men, teenage boys, and girls in settings where examination is not planned or is impractical (e.g., during preparticipation sports examinations or during initial medical evaluation of adolescent girls). Vaginal swabs—collected either by the health care provider at a pelvic examination or by the woman herself—are highly sensitive and specific for the diagnosis of chlamydial and gonococcal infection; they are now the preferred type of specimen for screening and diagnosis of these infections.

Although gonorrhea is now substantially less common than chlamydial infection in women in industrialized countries, screening tests for *N. gonorrhoeae* are still appropriate for women and teenage girls attending STD clinics and for sexually active teens and young women from areas of high gonorrhea prevalence. Multiplex NAATs that combine screening for *N. gonorrhoeae* and *C. trachomatis*—and, more recently, for *T. vaginalis*—in a single low-cost assay now facilitate the prevention and control of these infections for populations at high risk.

All patients who have newly detected STIs or are at high risk for STIs according to routine risk assessment as well as all pregnant

women should be encouraged to undergo serologic testing for syphilis and HIV infection, with appropriate HIV counseling before and after testing. Randomized trials have shown that risk-reduction counseling of patients with STIs significantly lowers subsequent risk of acquiring an STI; such counseling should now be considered a standard component of STI management. Preimmunization serologic testing for antibody to HBV is indicated for unvaccinated persons who are known to be at high risk, such as MSM and people who use injection drugs. In most young persons, however, it is more cost-effective to vaccinate against HBV without serologic screening. It is important to recognize that, while immunization against HBV has contributed to marked reductions in the incidence of infection with this virus, the majority of new cases that do occur are acquired through sex. In 2006, the CDC's Advisory Committee on Immunization Practices (ACIP) recommended the following: (1) Universal hepatitis B vaccination should be implemented for all unvaccinated adults in settings in which a high proportion of adults have risk factors for HBV infection (e.g., STD clinics, HIV testing and treatment facilities, drug-abuse treatment and prevention settings, health care settings targeting services to injection drug users or MSM, and correctional facilities). (2) In other primary care and specialty medical settings that provide care to adults at risk for HBV infection, health care providers should inform all patients about the health benefits of vaccination, the risk factors for HBV infection, and the persons for whom vaccination is recommended; they should vaccinate adults who report risk factors for HBV infection as well as any adult who requests protection from HBV infection. To promote vaccination in all settings, health care providers should implement standing orders to identify adults recommended for hepatitis B vaccination, should administer hepatitis B vaccine as part of routine clinical services, should not require acknowledgment of an HBV infection risk factor for adult vaccination, and should use available reimbursement mechanisms to remove financial barriers to hepatitis B vaccination.

In 2007, the ACIP made its first recommendation for routine immunization of 9- to 26-year-old girls and women with the quadrivalent HPV vaccine (against HPV types 6, 11, 16, and 18). In 2011, the ACIP recommended routine administration of quadrivalent HPV vaccine to boys at 11 or 12 years of age and to males 13–21 years of age who have not yet been vaccinated or who have not completed the three-dose vaccine series; HBV vaccination of men 22–26 years of age has also been recommended. Since that time, a nonavalent HPV vaccine has become available and has largely replaced the earlier vaccines. The optimal age for recommended vaccination is 11–12 years because of the very high risk of HPV infection after sexual debut.

Partner notification is the process of identifying and informing partners of infected patients about possible exposure to an STI and of examining, testing, vaccinating, and treating partners as appropriate. In a series of 22 reports concerning partner notification during the 1990s, index patients with gonorrhea or chlamydial infection named a mean of 0.75–1.6 partners, of whom one-fourth to one-third were infected; those with syphilis named 1.8–6.3 partners, with one-third to one-half infected; and those with HIV infection named 0.76–5.31 partners, with up to one-fourth infected. Persons who transmit infection or who have recently been infected and are still in the incubation period usually have no symptoms or only mild symptoms and seek medical attention only when notified of their exposure. Therefore, the clinician must encourage patients to participate in partner notification, must ensure that exposed persons are notified and treated, and must guarantee confidentiality to all involved. In the United States, local health departments often offer assistance in partner notification, treatment, and/or counseling. It seems both feasible and most useful to notify those partners exposed within the patient's likely period of infectiousness, which is often considered the preceding 1 month for gonorrhea, 1–2 months for chlamydial infection, and up to 3 months for early syphilis.

Persons with a new-onset STI always have a *source* contact who gave them the infection; in addition, they may have a *secondary* (*spread* or *exposed*) contact with whom they had sex after becoming infected.

The identification and treatment of these two types of contacts have different objectives. Treatment of the source contact (often a casual contact) benefits the community by preventing further transmission and benefits the source contact; treatment of the recently exposed secondary contact (typically a spouse or another steady sexual partner) prevents the development of serious complications (such as PID) in the partner, reinfection of the index patient, and further spread of infection. A survey of a random sample of U.S. physicians found that most instructed patients to abstain from sex during treatment, to use condoms, and to inform their sex partners after being diagnosed with gonorrhea, chlamydial infection, or syphilis; physicians sometimes gave the patients drugs for their partners. However, follow-up of the partners by physicians was infrequent. A randomized trial compared patients' delivery of therapy to partners exposed to gonorrhea or chlamydial infection with conventional notification and advice to partners to seek evaluation for STD; patients' delivery of partners' therapy, also known as *expedited partner therapy* (EPT), significantly reduced combined rates of reinfection of the index patient with *N. gonorrhoeae* or *C. trachomatis*. State-by-state variations in regulations governing this approach have not been well defined, but the 2015 CDC STD treatment guidelines describe its potential use. EPT, which is now commonly used by many practicing physicians, is currently permissible in 39 states and potentially allowable in another 8. (Updated information on the legal status of EPT is available at <http://www.cdc.gov/std/ept>.)

In summary, clinicians and public health agencies share responsibility for the prevention and control of STIs. In the current health care environment, the role of primary care clinicians has become increasingly important in STI prevention as well as in diagnosis and treatment, and the resurgence of bacterial STIs like syphilis and LGV among MSM—particularly those co-infected with HIV—emphasizes the need for risk assessment and routine screening.

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